Problem Statement

Imagine a university where everything is done on paper. Students have to fill out forms by hand to book a meeting with a teacher, reserve a conference room, or get permission for sports activities. They run around offices to submit these papers, wait days or weeks for approvals, and have no easy way to check if their request was accepted or rejected. Teachers and office staff spend a lot of time reviewing stacks of papers, which is slow and error-prone. Administrators struggle to keep track of records, generate reports, or manage user access securely. This old manual way wastes time, increases mistakes, and frustrates everyone—students, teachers, staff, and the university.

The problem is: The current manual system is inefficient, paper-heavy, hard to track, and not secure. It doesn't let people do things quickly online, and it doesn't handle different roles (like student vs. admin) well.

To fix this, we need a simple online system called the University Management and Permission System (or Circulation System). This web-based app will let users log in securely from any computer with internet. Students can submit requests online (like booking a teacher slot or a guest house room), track their status in real-time, and get notifications. Teachers and office authorities can review and approve/reject requests digitally. Admins can manage users, roles, and reports easily. The system will be user-friendly, role-based (so only authorized people see certain features), and secure to protect data.

In short, this system turns the messy paper process into a fast, digital one—saving time, reducing errors (aiming for less than 5% issues), and making university life smoother for everyone involved. It's built to be affordable, scalable for future updates, and compliant with university rules.

1.Inception

1.1 Introduction

Inception is the beginning phase of requirements engineering. It defines how does a software project get started and what is the scope and nature of the problem to be solved. The goal of the inception phase is to identify concurrence needs and conflict requirements among the stakeholders of a software project.

To establish the groundwork we have worked with the following factors related to the inception phases:

1. Identifying Stakeholders

2. Recognizing multiple viewpoints

3. Working towards collaboration

4. Asking the First Questions

1.2IdentifyingActors (Stakeholders):

- **Student:**The main user of the system. Students submit requests online (teacher appointments, sports field booking, complaints, conference room, etc.), track the status of their applications, and receive updates — all to save time and avoid running around with paper forms.
- **Teacher / Faculty Member:**Academic staff who receive and manage student requests (mainly teacher appointment requests). They can view available slots, approve, reject, or reschedule appointments easily from anywhere.
- **University:**The organization as a whole. It wants an efficient, low-cost, rule-compliant digital system that improves overall administrative work, reduces paperwork, and creates better records for the future.
- **Administrator :**The super-user of the system. Admins have full control — they create/delete users, assign/change roles, reset passwords, configure system settings, generate reports, and monitor everything for security and smooth operation.
- **Office Authority (SAO, Proctor, Register, Physical Education Officer):** Includes special roles like:SAO (Student Affairs Office), Proctor (handles studentdiscipline/complaints),Register (academic records office), Physical Education Officer (manages sports field/PE requests). They only see and handle

requests related to their own office. They review, approve, reject, or add comments on applications.

- **Guest House Manager:** The person responsible for guest house/room bookings. They check room availability, receive reservation requests from faculty/staff, approve or reject bookings, and keep proper guest records.
- **Developer:** The technical team/person who builds and maintains the system. They need clear requirements, easy-to-build structure, good security, scalability for future features, and a system that is not too difficult to update later.
- **Administrative staff:** Office helpers who support daily operations (may include data entry, helping users, or supporting office authorities). They benefit from faster processes and digital records (though they may not have very high-level access).

1.3Recognizing Multiple viewpoints:

1. Student Viewpoints

- Easy to use and user-friendly system
- Access the system via the Internet
- Easy submission of applications and requests
- Track application status online
- Save time and reduce paperwork

2. Teacher / Faculty Member Viewpoints

- Easy to review student applications
- Approve or reject requests online
- Role-based access to system features
- Fast communication with students and administration

3. University Viewpoints

- System should be within budget
- Improve efficiency of academic and administrative work
- Maintain proper digital records
- No violation of university rules and regulations

4. Administrator Viewpoints

- Full control over the system
- Manage users and roles easily
- Secure authentication system
- Generate reports and monitor system activities
- Easy maintenance and updates

5. Office Authority (SAO, Proctor, Physical Education Officer) Viewpoints

- Access applications related to their office only
- Approve, reject, or comment on requests
- Easy record keeping
- Secure and role-based access

6. Guest House Manager Viewpoints

- Manage room booking requests easily
- Check room availability online
- Approve or reject guest accommodation requests
- Maintain guest records efficiently

7. Developer Viewpoints

- Clear and unambiguous requirements
- Easy to develop and maintain
- Scalable system for future updates
- Secure and reliable architecture

1.4 Working towards collaboration

Every stakeholder has their own requirement. In this step, we merged these requirements. We followed following steps to complete the task:

- Identify the common and conflicting requirements
- Categorize the requirements
- Take priority points for each requirements from stakeholders and on the basis of this voting prioritize the requirements
- Make final decision about the requirements.

Common requirements:

- User friendly and efficient system
- Easy to operate
- The application can be accessed from any computer that has Internet access.

Conflicting Requirements:

- Minimum maintenance cost
- Availability of all requirements within the budget
- Easy access
- Restrict access to functionality of the system based upon user roles
- No ambiguous requirement

Final Requirements:

- Error-free system (Maximum 5% error may be acceptable).
- The system must be accessible via the Internet.
- Allow valid users to log in and log out securely.
- Restrict access to system features based on user roles (Student, Teacher, Office Authority, Admin, etc.).
- Allow administrators to manage users, change roles, and configure system settings.
- Allow students to submit applications and requests online.
- Allow users to track the status of their applications.
- Allow teachers and office authorities to approve or reject requests online.
- Allow administrators to generate reports on applications, approvals, and system usage.
- The application should be installed and maintained on a single server.
- Ensure data security and proper record management.

1.5Asking the First Questions

We set our first set of context-free questions focuses on the customer and other stakeholders, overall project goals and benefits. The questions are mentioned above. These questions helped us to identify all stakeholders, measurable benefit of the successful implementation and possible alternatives to custom software development. Next set of question helped us to gain a better understanding of problem and allows the customer to voice his or her perception about the solution. The final set of question focused on the effectiveness of the communication activity itself.

1.6 Conclusion:

The inception phase helped us to build a basic understanding of the university management and permission system.It helped identify the stakeholders who will benefit from the system, such as students, faculty members, office authorities, and administrators.This phase also helped define the nature of the software and establish effective communication with stakeholders for successful system development.

2.Elicitation

2.1 Introduction:

Elicitation is a process that helps stakeholders clearly define what the system should do. During this step, we faced common problems such as unclear requirements, changing needs, and misunderstanding between users and developers.

To overcome these issues, we followed an organized and systematic elicitation process. This helped us clearly understand user needs and define correct system requirements.

2.2Eliciting Requirements:

Unlike the inception phase where simple question–answer methods were used, the elicitation phase focuses on deeper understanding.This phase combines problem solving, discussion, negotiation, and requirement specification.

To elicit requirements for our system, we completed the following four activities:

- Collaborative Requirements Gathering
- Quality Function Deployment (QFD)
- Usage Scenarios
- Elicitation Work Products

2.3 Collaborative Requirements Gathering:

Different collaborative techniques were used to collect requirements from stakeholders. We followed the steps below:

- Meetings were conducted with students, teachers, office authorities, and administrators of the university.
- Stakeholders were asked about problems in the existing manual process.

- They shared their expectations for an online and automated system.
- Finally, we prepared the final list of system requirements based on these discussions.

2.4 Quality Function Deployment (QFD):

Quality Function Deployment helps convert customer needs into technical system requirements.
It focuses on increasing user satisfaction through better software design.

Using QFD, we identified the following requirements for our system.

Normal Requirements:

These requirements were directly stated by stakeholders during meetings:

- The system should be accessible via the Internet.
- Users should be able to log in and log out securely.
- Role-based access for students, teachers, office authorities, and administrators.
- Students should be able to submit applications and requests online.
- Teachers and office authorities should be able to approve or reject requests.
- Users should be able to track the status of their applications.
- Administrators should be able to manage users and system settings.
- The system should generate reports on applications and approvals.
- The system should be easy to use and user friendly.
- A user guide should be provided for system usage.

Expected Requirements:

These requirements are basic and expected, even if not clearly stated by users:

- Maintain a database of all users and applications.
- Allow administrators to change user roles.
- Allow administrators to reset user passwords.
- Ensure secure authentication and authorization.
- Automatically update application status after approval or rejection.
- Maintain proper records of approvals and rejections.
- Allow administrators to configure system parameters.
- Ensure data consistency and reliability.
- Provide error handling for invalid actions.

- Ensure system availability and performance.

Exciting Requirements:

These features go beyond basic expectations and increase user satisfaction:

- Automatic email or notification updates on application status.
- Simple and modern web-based user interface.
- Mobile-friendly system design.
- Online help and tooltips for users.
- Secure login using two-factor authentication (optional).

2.5 Usage Scenarios:

- First, a user creates an account or logs into the system using valid credentials. After logging in, the user selects the required service or application type.
- A student submits an application online.
- The system stores the application and updates its status as “Pending”.
- The application is forwarded to the concerned teacher or office authority.
- The authority reviews the application and approves or rejects it.
- The system updates the status automatically.
- The student can track the application status online.

In case of rejection, comments or reasons are shown to the user.

2.6 Elicitation Work Product:

The outputs of the elicitation phase include:

- A clear statement of requirements for the university management and permission system.
- A defined scope of the system.
- A list of stakeholders involved in requirement elicitation.
- Set of usage scenarios.
- Description of the system’s technical environment

3.Use Cases

3.1 Use Case Scenario:

Level 0	Level 1	Level 2	Primary Actor
Circulation System	Authentication	Sign Up	Student, Faculty
		Sign In	Administrator, Student, Faculty, Office Authority, Guest House Manager
		Sign Out	All users
		Change Pass	All users
	Teacher Appointment	View Teacher Available Slots	Student
		Request Appointment	Student
		Approve/Reject/Reschedule Appointment	Faculty/Teacher
	Physical Education Approvals	View Field Available Slots	Student
		Submit or Cancel Request	Student
		Approve/Reject Field Request	Physical Education Officer
	Proctor Communication	Raise Complaint/Issue to Proctor	Student
		View & Respond to Student Issues	Proctor
		Mark Issue as Resolved	Proctor
	Conference Room Booking	View Room Availability Calendar	Student, Faculty
		Submit or Cancel Request	Student, Faculty

		Approve/Reject Room Booking	Office Authority
Circulation System	Guest House Reservation	View Guest House Room Availability	Faculty, Staff
		Submit Guest House Room Reservation Request	Faculty, Staff
		Approve/Reject Guest House Room Reservation	Guest House Manager
	User & System Management	Manage Users (Create, Update, Deactivate)	Admin
		Assign/Change User Roles	Admin
		Configure System Settings (Notifications, etc.)	Admin

3.2 Use Case Description:

3.2.1. Authentication

i.Use case: Sign Up

Primary Actors: Student, Faculty

Goal in context: To register a new user into the system

Precondition:

- System must support user registration
- Registration interface must be available

Triggers: User needs to create an account

Scenario:

- User visits the registration page
- Enters required information
- System validates input
- System creates a new user account
- Confirmation message is shown

Exception:

- Invalid input
- Username already exists

Priority: Essential, must be implemented
When Available: First increment

ii.Use case: Sign In
Primary Actors: Administrator, Student, Faculty, Office Authority, Guest House Manager
Goal in context: To access the system
Precondition: User must be registered

Triggers: User wants to enter the system

Scenario:

- User visits login page
- Enters username and password
- System verifies credentials
- User is redirected to dashboard

Exception:

- Invalid username
- Incorrect password
- User blocked

Priority: Essential, must be implemented
When Available: First increment

iii.Use case: Sign Out
Primary Actors: All Users
Goal in context: To safely exit the system
Precondition: User must be logged in

Triggers: User wants to log out

Scenario:

- User clicks logout button
- System ends the session

Priority: Essential
When Available: First increment

iv.Use case: Change Password
Primary Actors: All Users
Goal in context: To update account password
Precondition: User must be logged in

Triggers: User wants to change password

Scenario:

- User opens profile settings
- Enters old password
- Enters new password
- System updates password

Exception:

- Weak password
- Old password incorrect

Priority: Essential
When Available: First increment

3.2.2. Teacher Appointment

i.Use case:View Teacher Available Slots
Primary Actors: Student
Goal in context: To view available appointment slots of teachers
Precondition**:** Student must be logged in

Triggers: Student wants to book an appointment

Scenario:

- Student logs in
- Selects teacher appointment option
- Views available time slots

Exception:No slots available

Priority: Essential
When Available: First increment

ii.Use case:Approve / Reject / Reschedule Appointment
Primary Actors: Faculty / Teacher

Goal in context: To manage appointment requests
Precondition: Appointment request must exist

Triggers: New appointment request

Scenario:

- Teacher logs in
- Views appointment requests
- Approves, rejects, or reschedules appointment
- System updates appointment status

Exception:Unauthorized access

Priority: Essential
When Available: First increment

3.2.3 Physical Education Approvals

i.Use case:Submit or Cancel PE Request
Primary Actors: Student
Goal in context: To submit or cancel physical education request
Precondition: Student must be logged in

Triggers: Student wants PE approval

Scenario:

- Student logs in
- Submits or cancels PE request
- System records request

Exception:Invalid request

Priority: Essential
When Available: First increment

ii.Use case:Approve / Reject PE Request
Primary Actors: Physical Education Officer
Goal in context: To approve or reject PE requests
Precondition: PE request must exist

Triggers: New PE request submitted

Scenario:

- Officer logs in
- Reviews PE request
- Approves or rejects request
- System updates request status

Exception:Unauthorized access

Priority: Essential
When Available: First increment

3.2.4. Proctor Communication

i.Use case:Raise Complaint / Issue to Proctor
Primary Actors: Student
Goal in context: To report issues to proctor
Precondition: Student must be logged in

Triggers: Student faces an issue

Scenario:

- Student logs in
- Submits complaint details
- System forwards issue to proctor

Exception:Invalid input

Priority: Essential
When Available: First increment

ii.Use case:View & Respond to Student Issues
Primary Actors: Proctor
Goal in context: To respond to student complaints
Precondition: Proctor must be logged in

Triggers: New complaint submitted

Scenario:

- Proctor logs in
- Views complaints
- Responds to student issue

Exception:Issue not found

Priority: Essential
When Available: First increment

iii.Use case:Mark Issue as Resolved
Primary Actors: Proctor
Goal in context: To close resolved issues
Precondition: Issue must be reviewed

Triggers: Issue resolved

Scenario:

- Proctor marks issue as resolved
- System updates status

Priority: Essential
When Available: First increment

3.2.5.Conference Room Booking

i.Use case:View Room Availability Calendar
Primary Actors: Student, Faculty, Admin
Goal in context: To check available conference rooms
Precondition: User must be logged in

Triggers: Need to book a conference room

Scenario:

- User logs in
- Opens room availability calendar
- Views available time slots

Exception:Calendar loading failure

Priority: Essential
When Available: First increment

ii.Use case:Submit or Cancel Room Booking Request
Primary Actors: Student, Faculty
Goal in context: To request or cancel a conference room booking
Precondition: Room availability must be checked

Triggers: User wants to reserve or cancel a room

Scenario:

- User selects room and time
- Submits booking request OR cancels request
- System updates booking status

Exception:

- Slot already booked
- Invalid date/time

Priority: Essential
When Available: First increment

iii.Use case:Approve / Reject Room Booking
Primary Actors: Office Authority
Goal in context: To approve or reject room booking requests
Precondition: Pending booking request exists

Triggers: New booking request

Scenario:

- Office Authority logs in
- Reviews booking request
- Approves or rejects request
- System notifies requester

Exception:Unauthorized action

Priority: Essential
When Available: First increment

3.2.6. Guest House Reservation

i.Use case:View Guest House Availability
Primary Actors: Faculty, Staff
Goal in context: To check guest house availability
Precondition: User must be logged in

Triggers: Need to reserve guest house

Scenario:

- User logs in
- Views guest house availability
- Selects preferred dates

Exception:Data not available

Priority: Essential
When Available: First increment

ii.Use case: Submit Guest House Reservation Request
Primary Actors: Faculty, Staff
Goal in context: To request guest house accommodation
Precondition: Availability confirmed

Triggers: Need for guest accommodation

Scenario:

- User submits reservation request
- System forwards request to manager

Exception:

- Invalid date
- Request submission failure

Priority: Essential
When Available: First increment

iii.Use case:Approve / Reject Guest House Reservation
Primary Actors: Guest House Manager
Goal in context: To manage guest house reservations
Precondition: Reservation request exists

Triggers: New reservation request

Scenario:

- Manager logs in
- Reviews request
- Approves or rejects reservation
- System updates reservation status

Exception:Unauthorized access

Priority: Essential
When Available: First increment

3.2.7. User & System Management

i.Use case: Manage Users (Create, Update, Deactivate)
Primary Actors: Admin
Goal in context: To manage system users
Precondition: Admin must be logged in

Triggers: User management required

Scenario:

- Admin selects user management
- Creates, updates, or deactivates users
- System saves changes

Exception:Invalid user data

Priority: Essential
When Available: First increment

ii.Use case: Assign / Change User Roles
Primary Actors: Admin
Goal in context: To control user access levels
Precondition: User must exist

Triggers: Role update needed

Scenario:

- Admin selects user
- Assigns or changes role
- System updates role

Exception:Unauthorized role change

Priority: Essential
When Available: First increment

iii.Use case: Configure System Settings
Primary Actors: Admin
Goal in context: To configure system notifications and settings
Precondition: Admin login required

Triggers: System configuration required

Scenario:

- Admin accesses system settings

- Updates configurations
- System applies changes

Exception:Configuration error

Priority: Essential
When Available: First increment

4.Use case Diagram

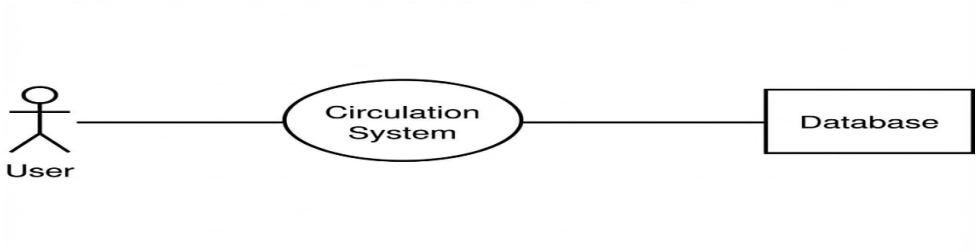

Fig: Level 0 For Circulation System

Level 0 Use Case Diagram: Shows the most basic interaction where a general user interacts with the Circulation System, which connects to a database.

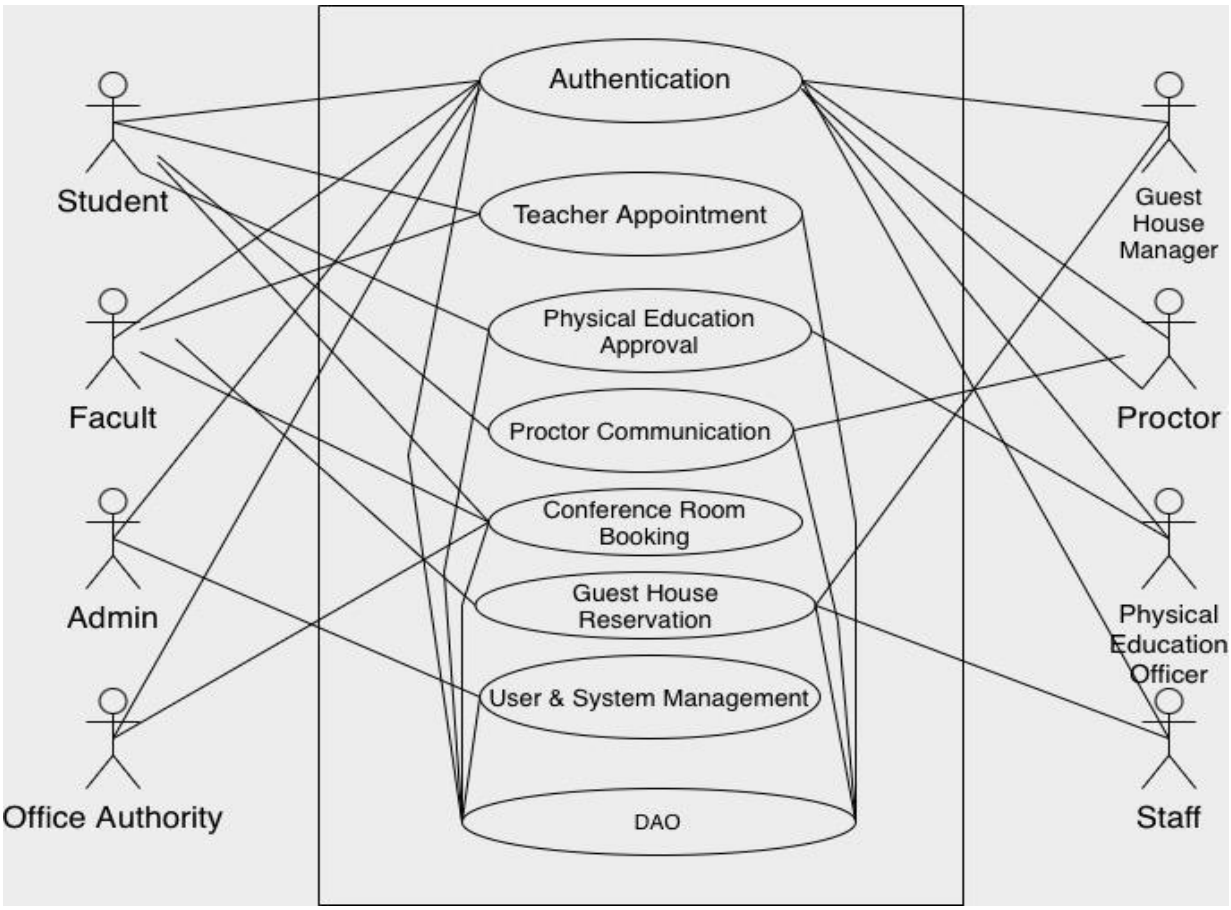

Fig: Level 1 For Circulation System.

Level 1 Use Case Diagram: Provides a high-level view of all majorsystem functions, such as Authentication, Teacher Appointments, and Room Bookings, and the various users (Student, Faculty, Admin, etc.) who use them.

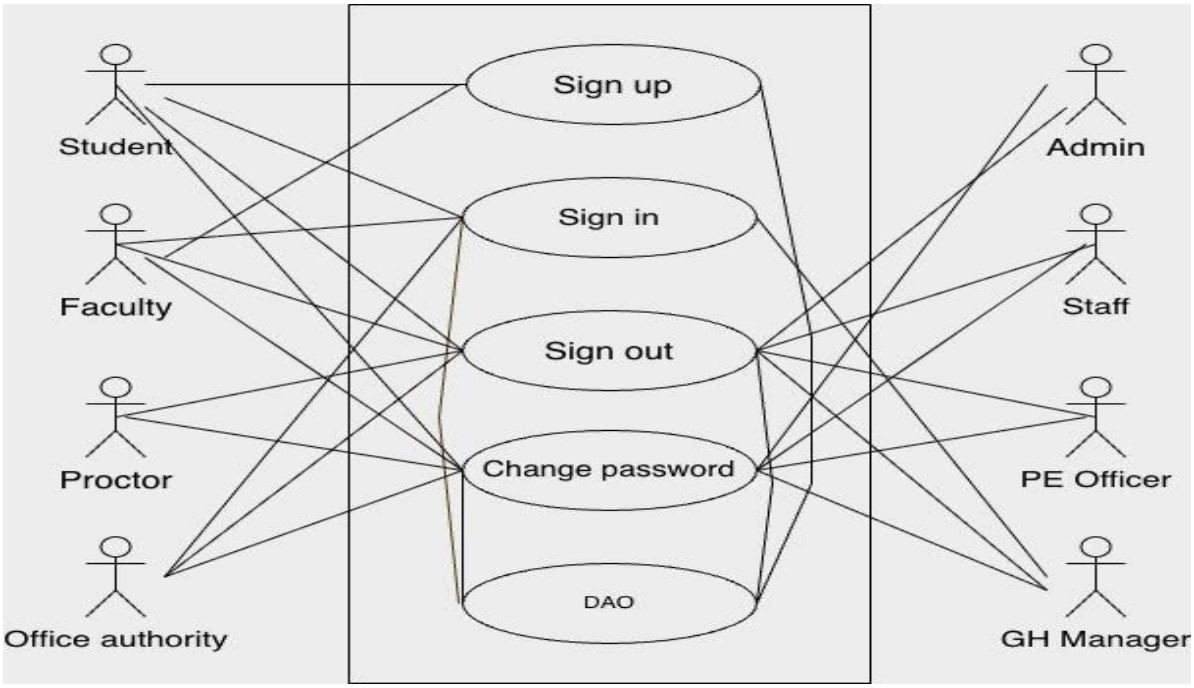

Fig: Level 2.1(Authentication) for circulation system

Authentication (Level 2.1): Outlines the core security actions available to all users, including signing up, signing in, signing out, and changing passwords.

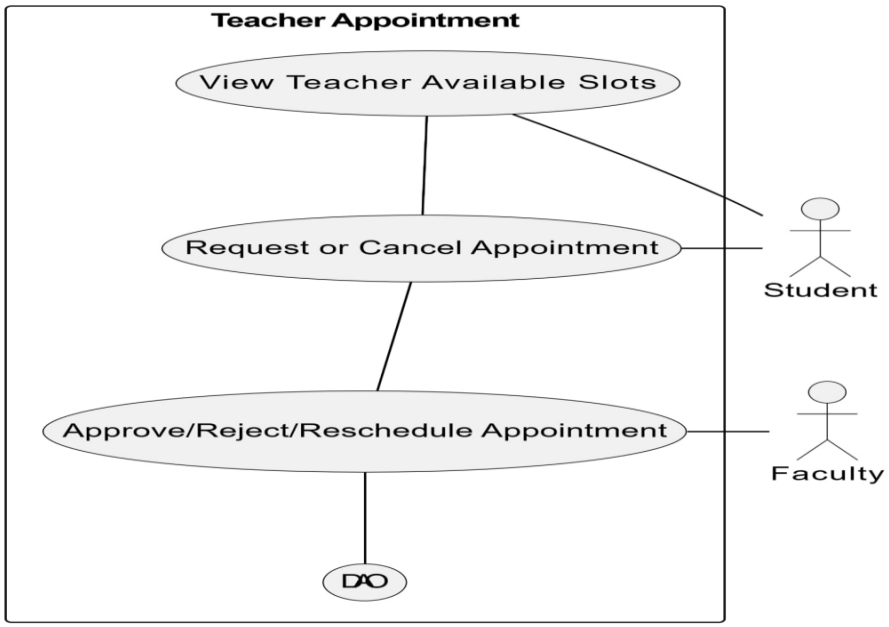

Fig: Level 2.2(Teacher Appointment) for circulation system

Teacher Appointment (Level 2.2): Shows how students can request appointments and how faculty members can approve, reject, or reschedule them.

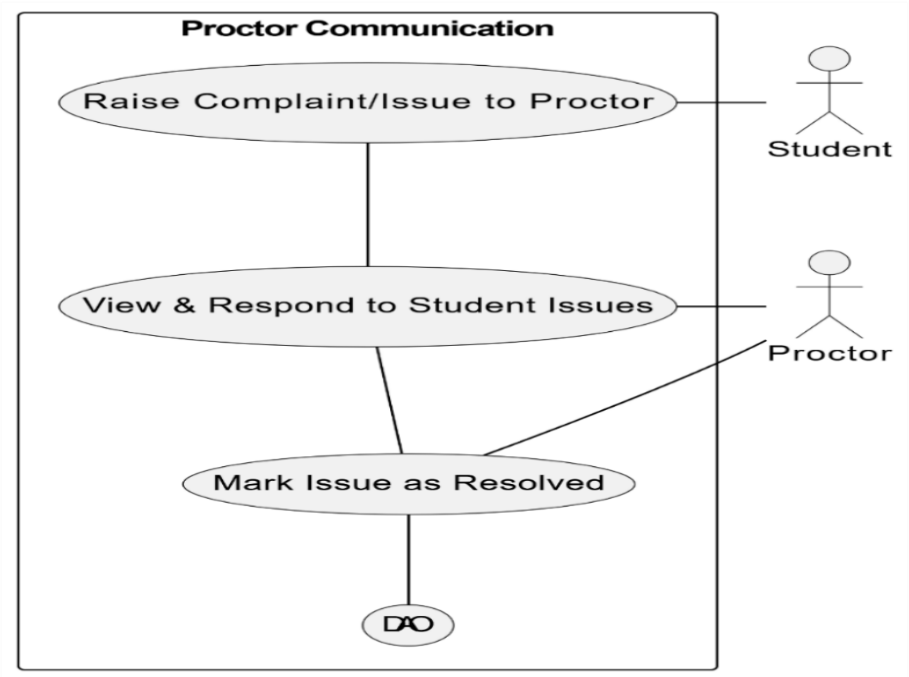

Fig: Level 2.3(Proctor Communication) for circulation system

Proctor Communication (Level 2.3): Defines the interaction where students raise complaints or issues and the Proctor views and resolves them.

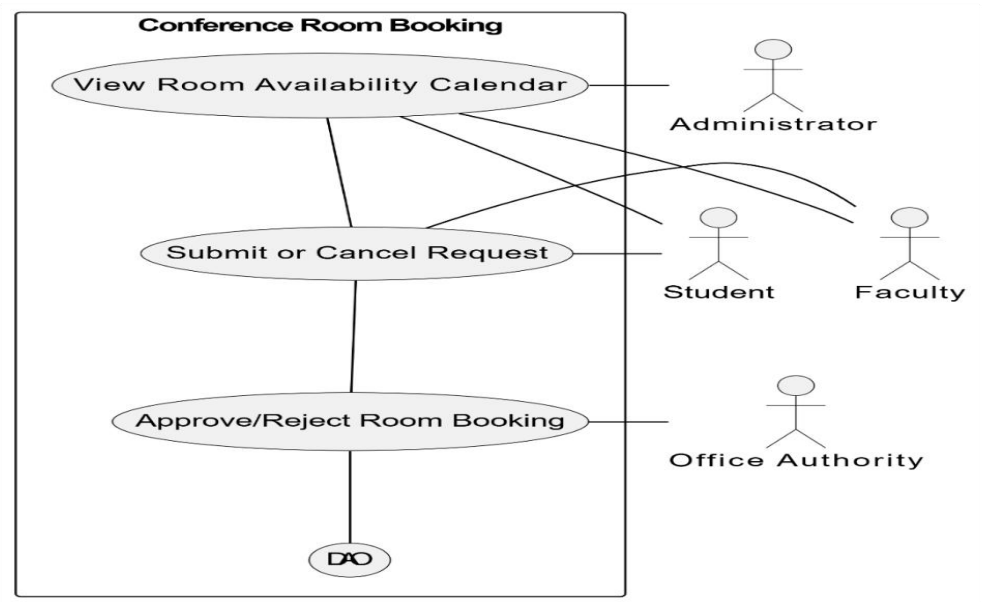

Fig: Level 2.4(Conference room booking) for circulationsystem

Proctor Communication (Level 2.4): Defines the interaction where students raise complaints or issues and the Proctor views and resolves them.

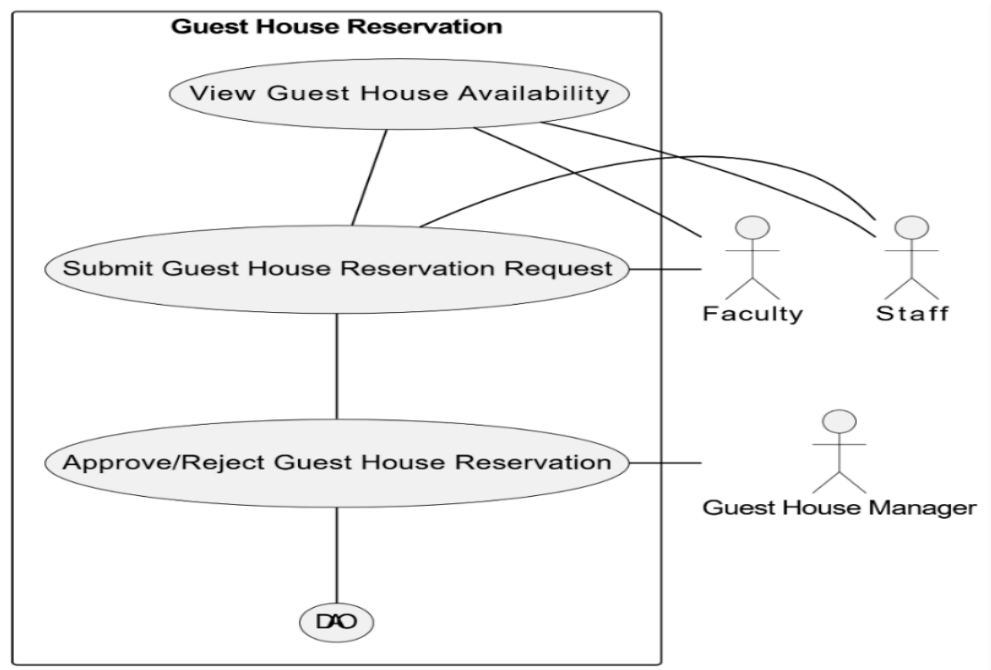

Fig:Level 2.5(Guest house reservation) for circulation system

Guest House Reservation (Level 2.5): Illustrates the process for Faculty or Staff to request housing, which is managed and approved by the Guest House Manager

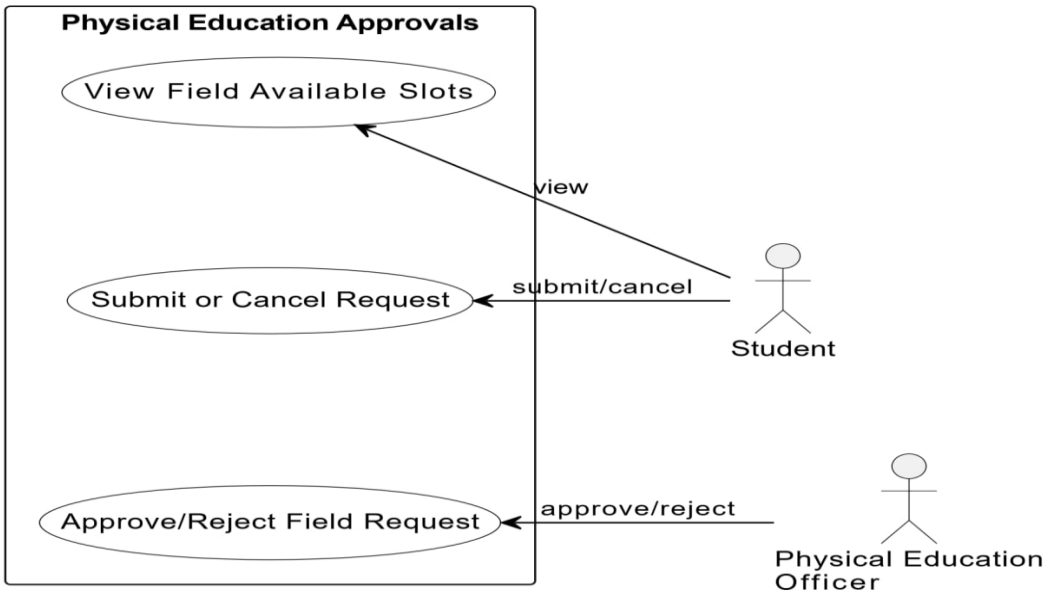

Fig: Level 2.6(Physical Education Approvals) for circulation system

Physical Education Approvals (Level 2.6): Shows students requesting the use of a sports field and the Physical Education Officer managing those approvals.

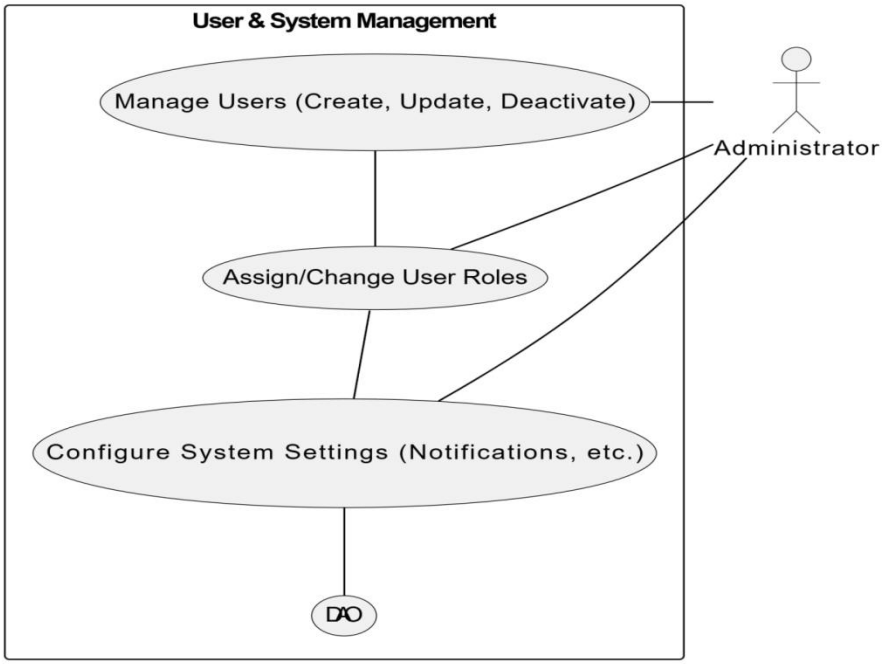

Fig: Level 2.7(User & System Management) for circulation system

User & System Management (Level 2.7): Shows the Administrator's tasks, such as managing user accounts, assigning roles, and configuring system-wide settings

5.Scenario Based Modeling

Use Case1: Sign up

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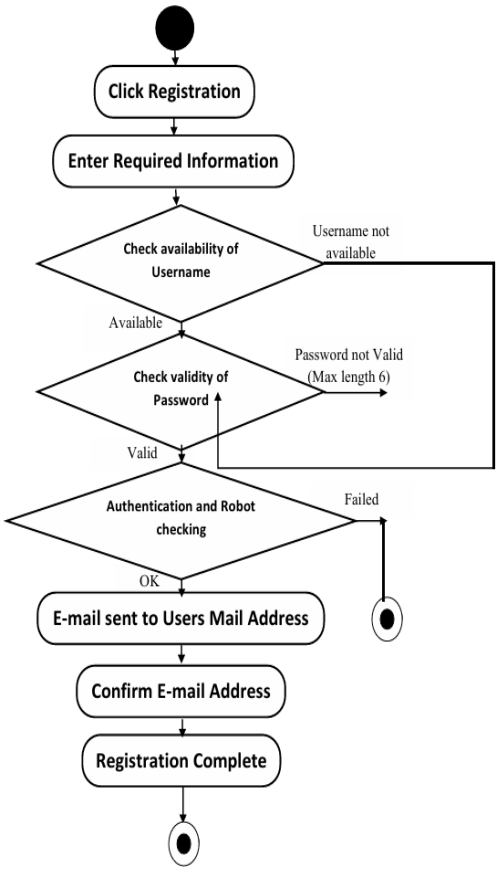


Fig: Activity Diagram(Sign Up)

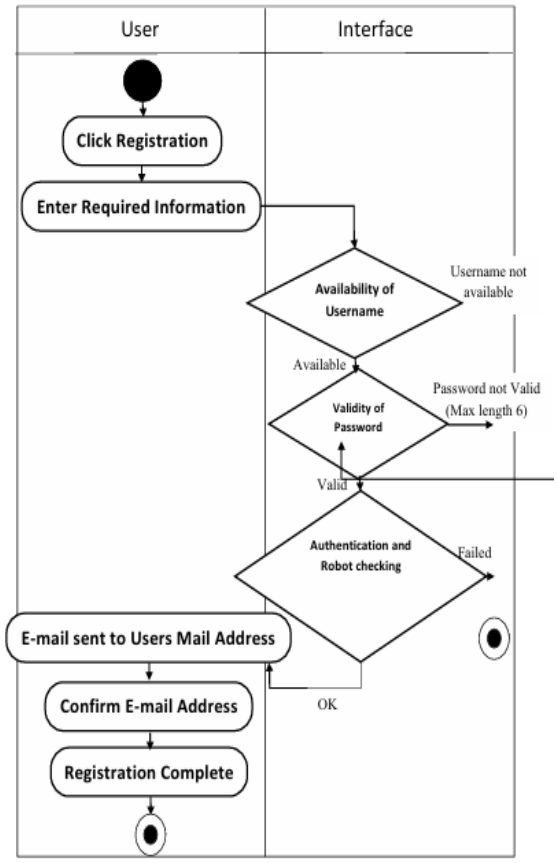


Fig: Swimlane Diagram(Sign Up)

Use case2:Sign in

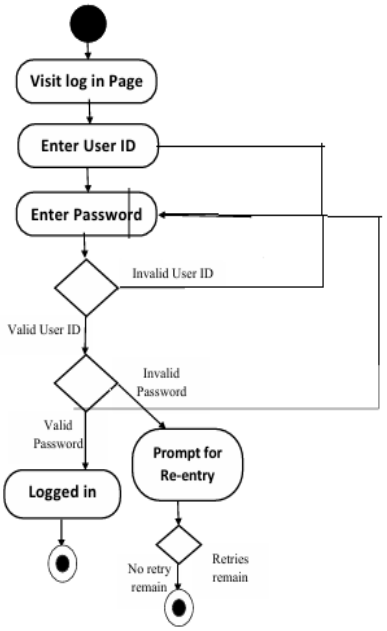


Fig: Active Diagram(Sign in)

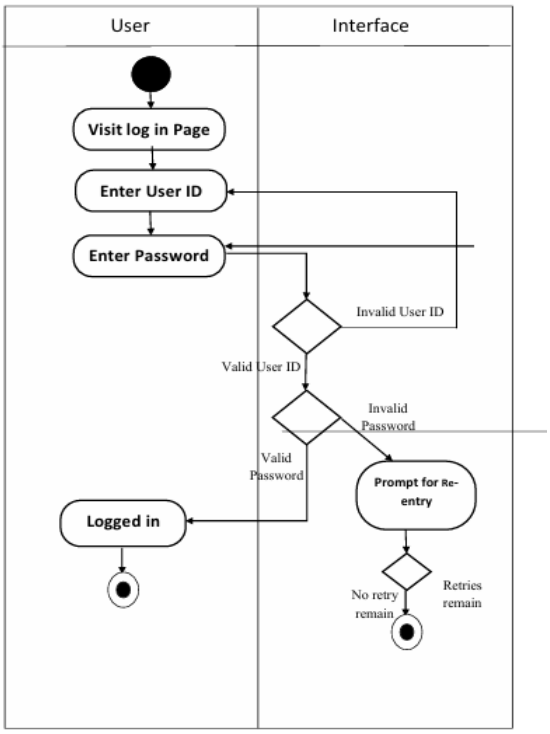


Fig: Swimlane Diagram(Sign in)

Use Case3: Sign out

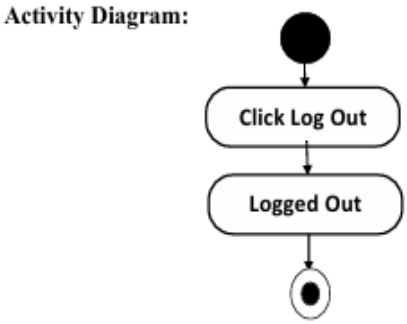


Fig: Activity Diagram(Sign Out)

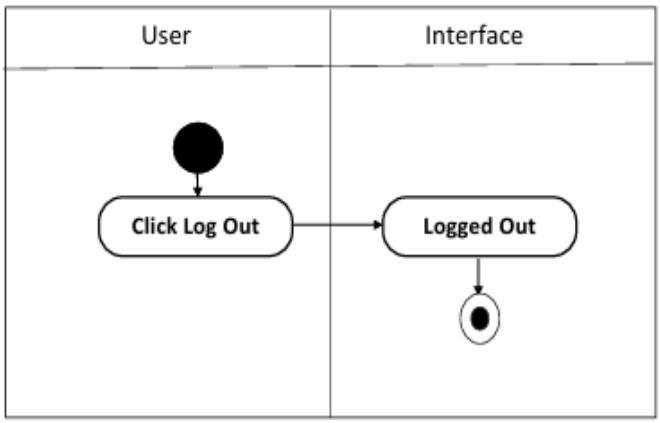


Fig: Swimlane Diagram (Sign Out)

Use Case4: Log in

S

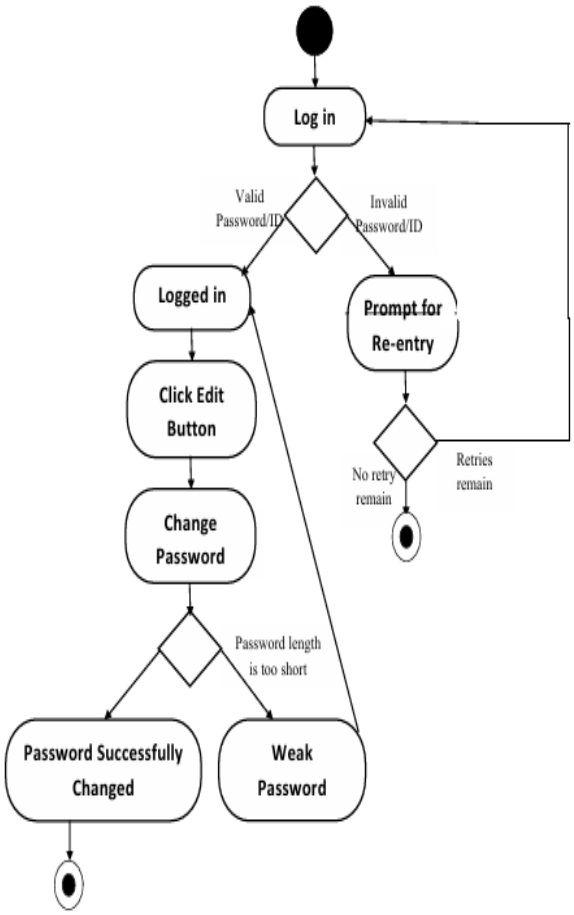


Fig:ActiveDiagram(Log In)

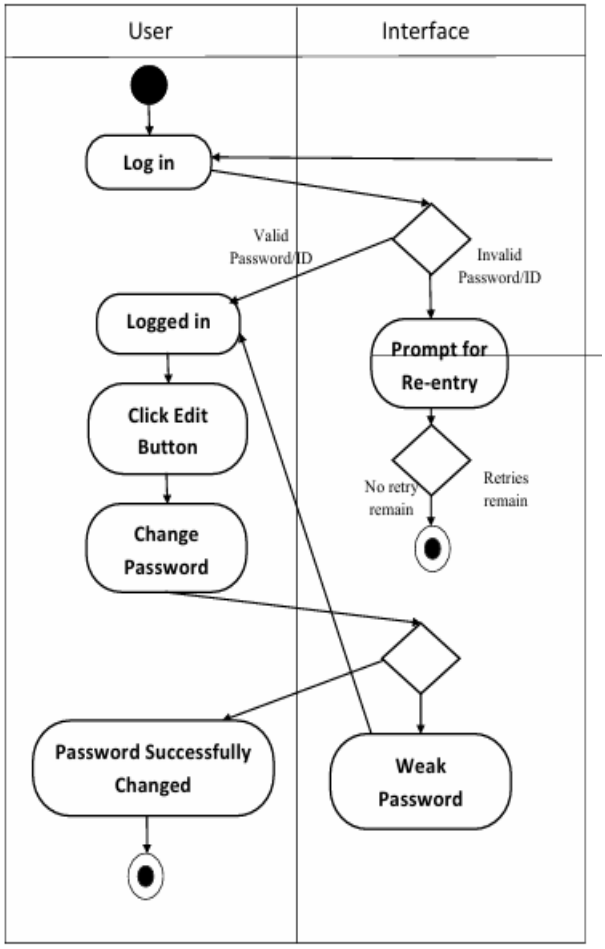


Fig: Swimlane Diagram(Log In)

Use case5 : View Teacher Slot

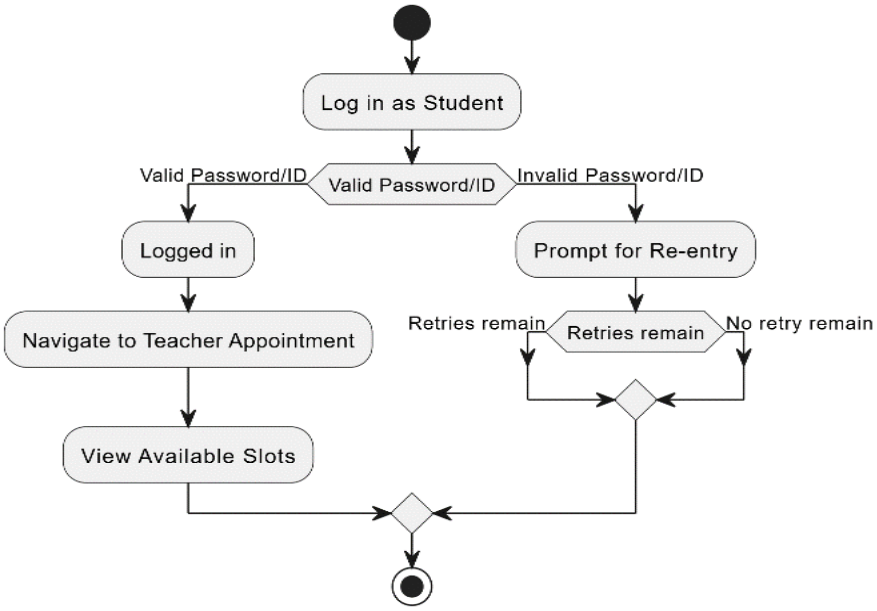


Fig: Activity Diagram(View Teacher Slot)

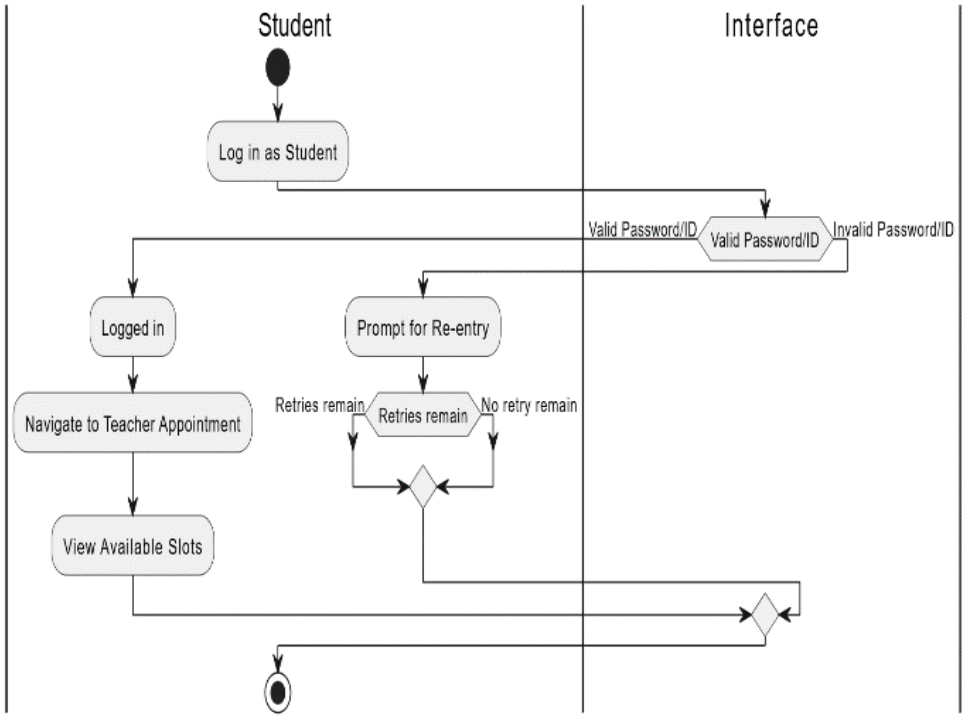


Fig:SwimlaneDiagram(View Teacher Slot)

Use case 6: Request Appointment

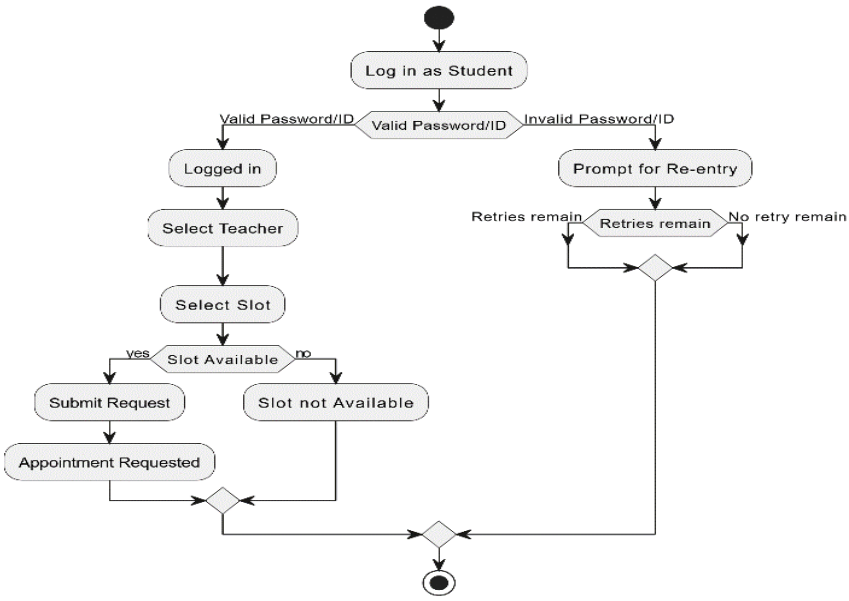


Fig: Activity Diagram(Request Appointment)

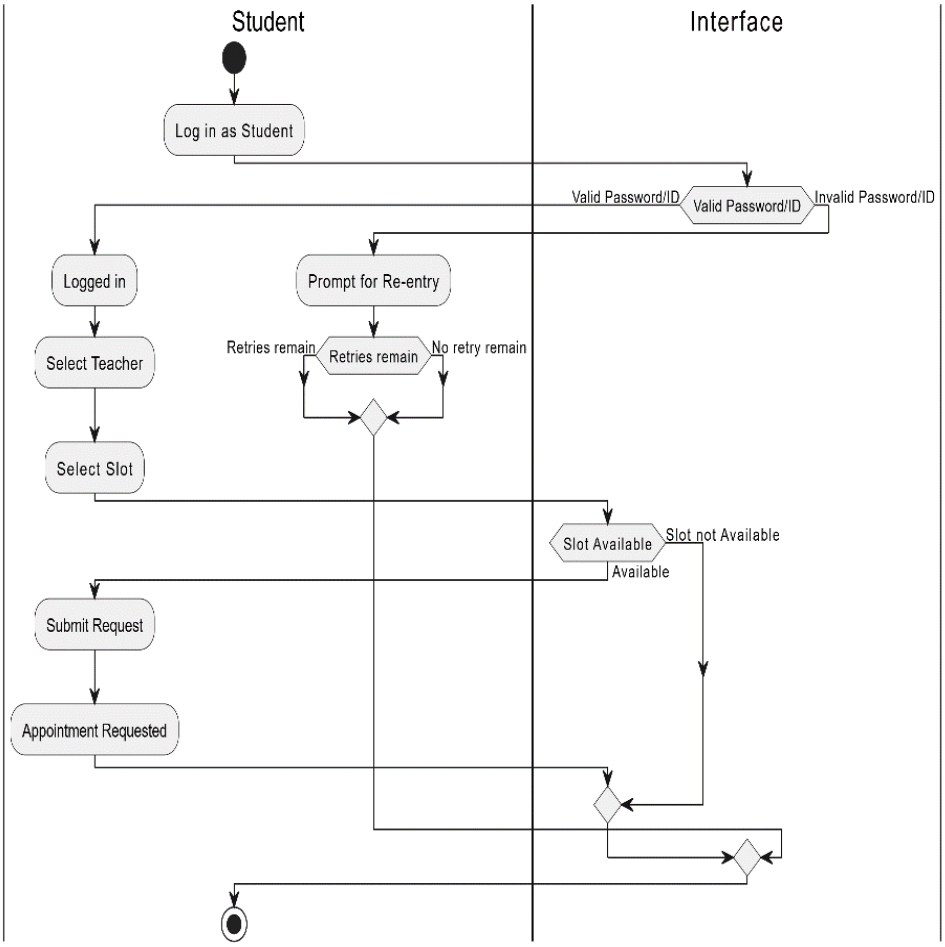


Fig: SwimlaneDiagram(Request Appointment)

Use case 7: Approve / Reject / Reschedule

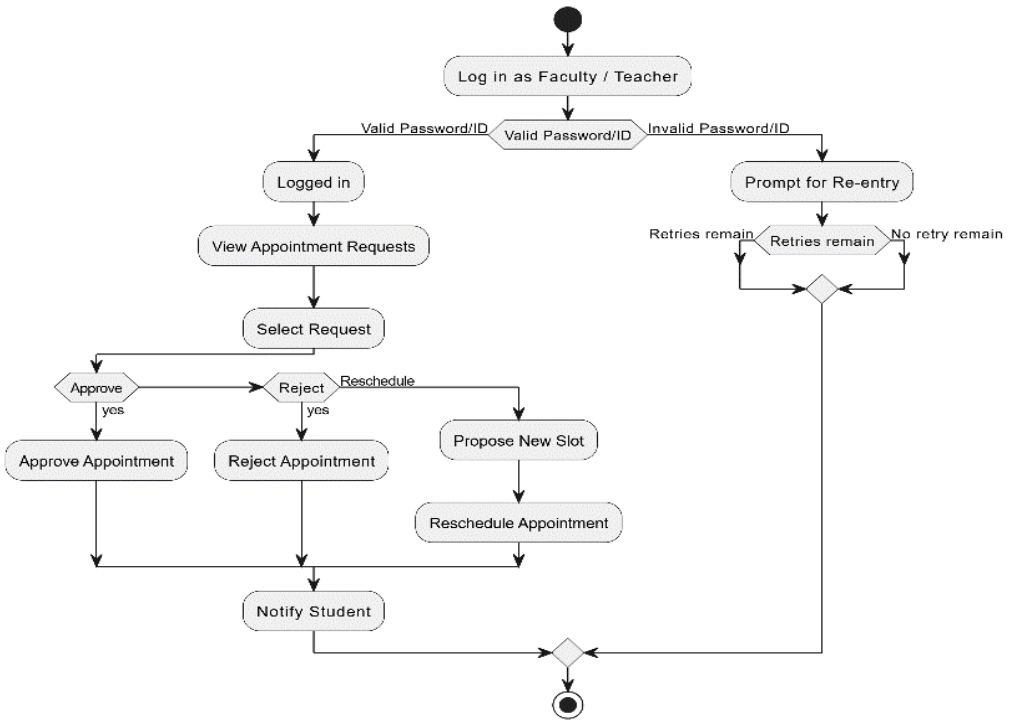


Fig: Activity Diagram(Approve / Reject / Reschedule)

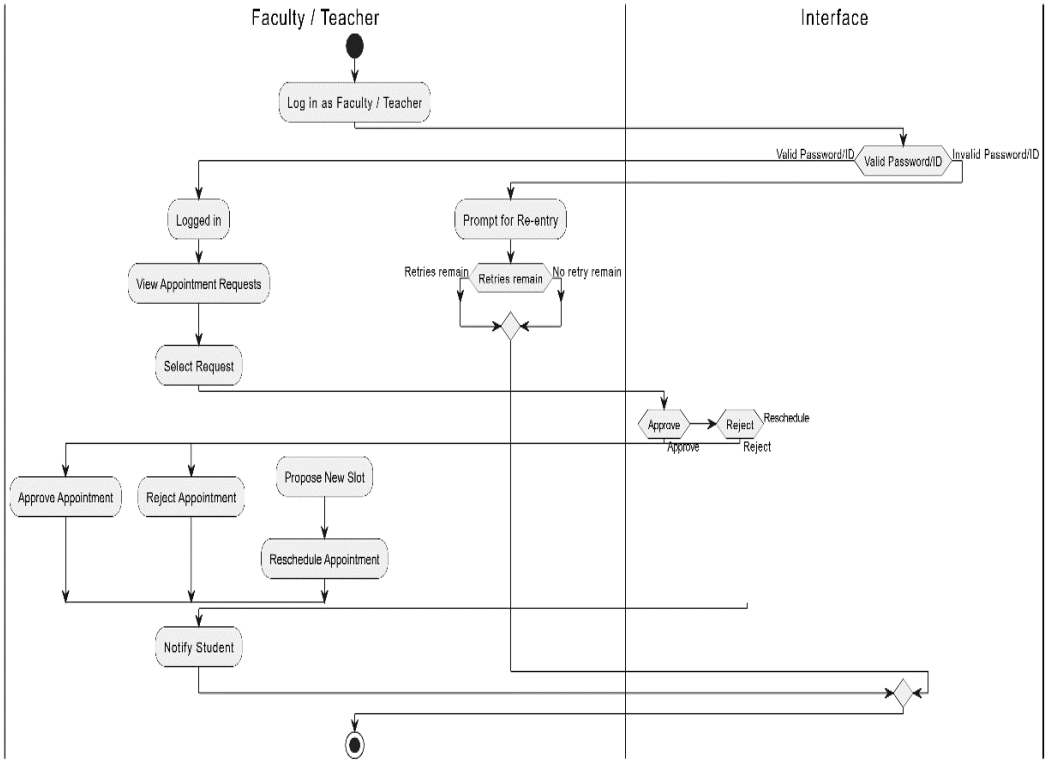


Fig: SwimlaneDiagram(Approve / Reject / Reschedule)

Use case 8: View Field Available Slots

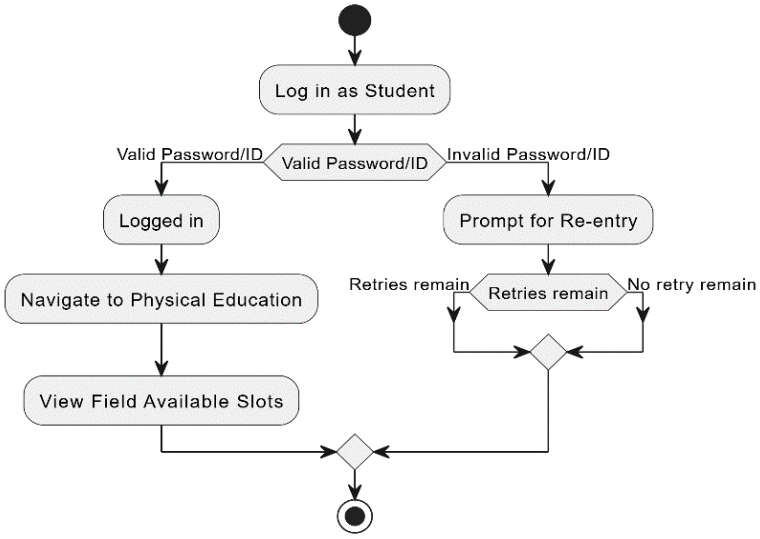


Fig: Activity Diagram(View Field Available Slots)

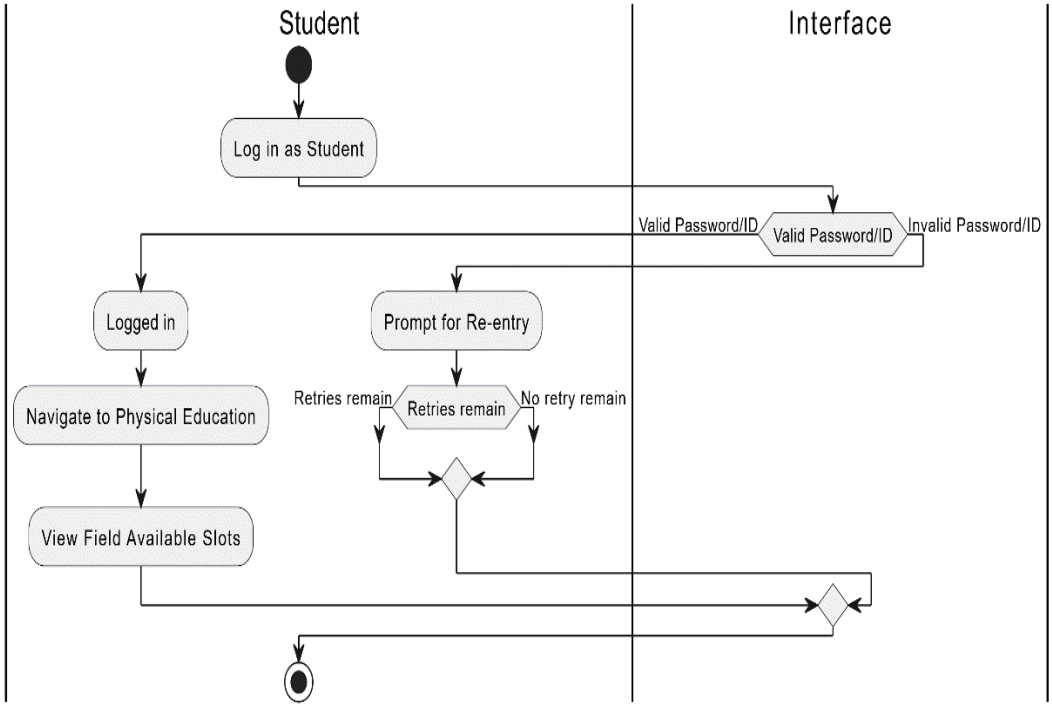


Fig: SwimlaneDiagram(View Field Available Slots)

Use case 9: Submit or Cancel Request

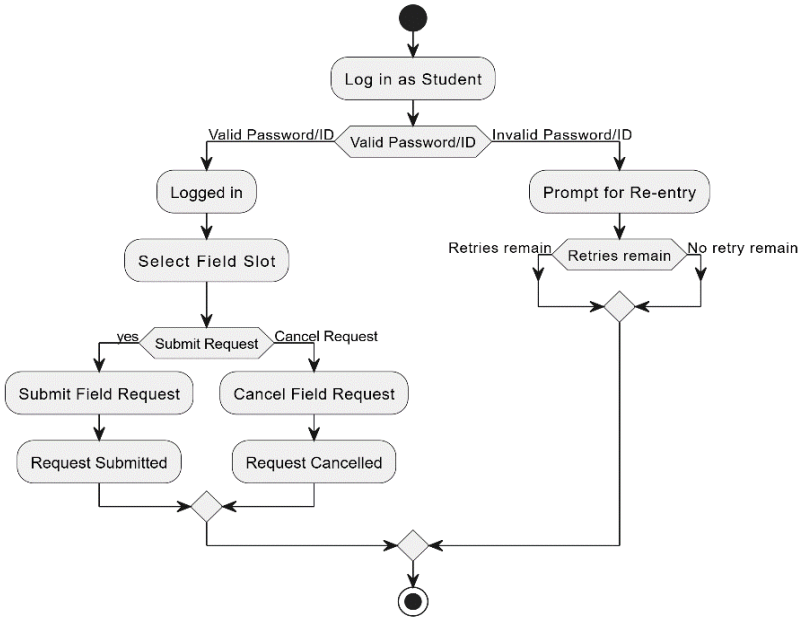


Fig: Activity Diagram(Submit or Cancel Request)

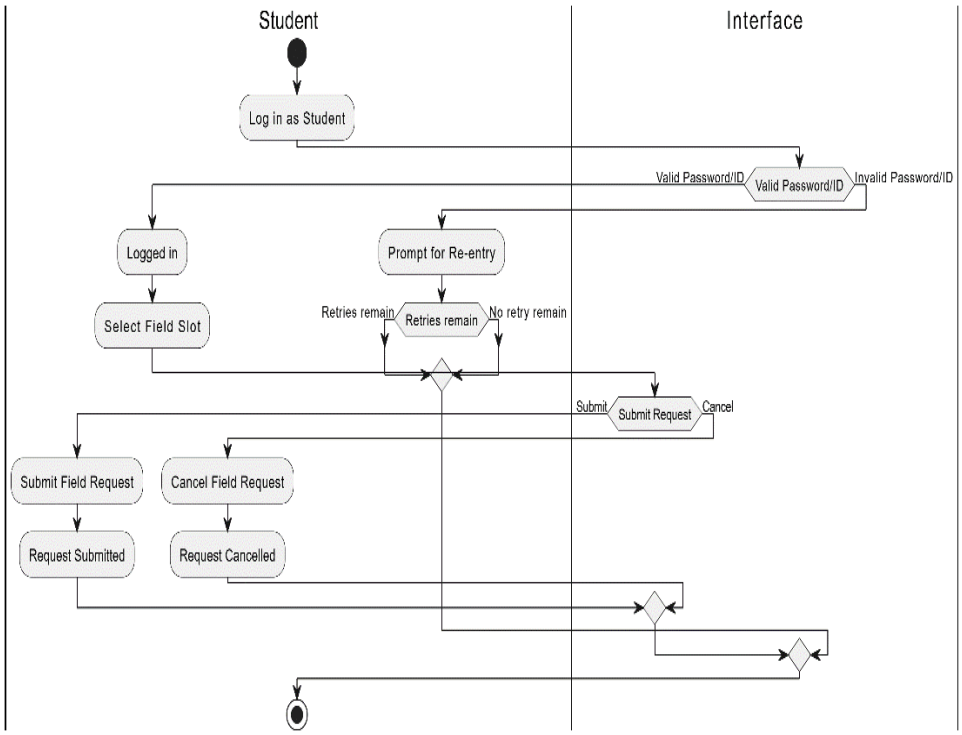


Fig: SwimlaneDiagram(Submit or Cancel Request)

Use case 10: Approve / Reject Field Request

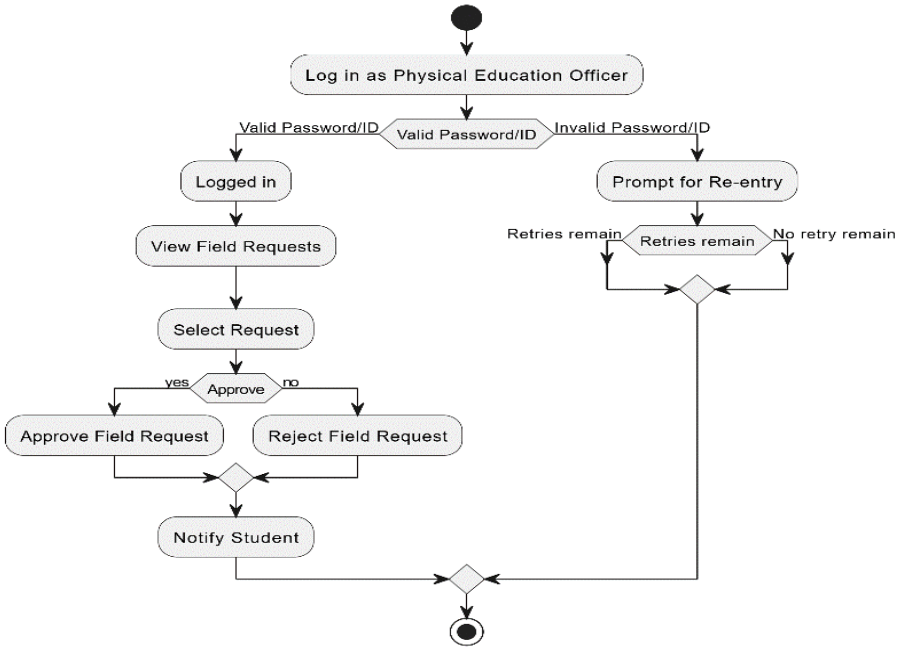


Fig: Activity Diagram(Approve / Reject Field Request)

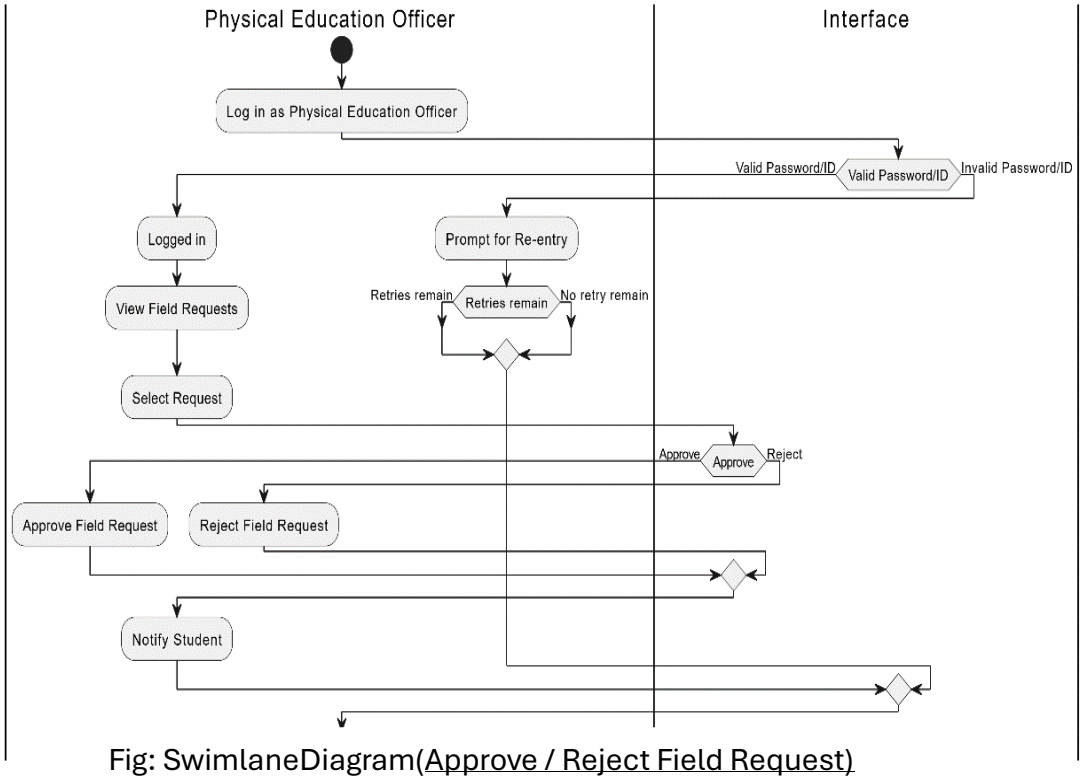


Fig: SwimlaneDiagram(Approve / Reject Field Request)

Use case 11: Raise Complaint / Issue

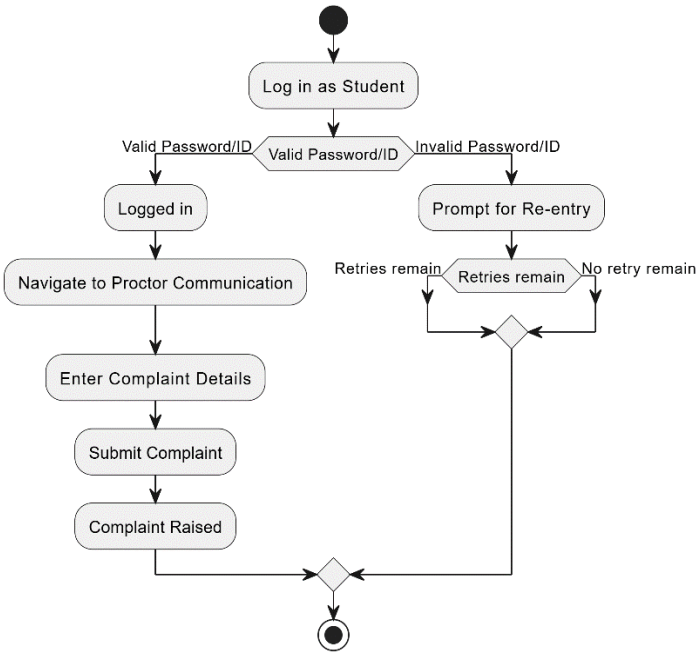


Fig: Activity Diagram(Raise Complaint / Issue)

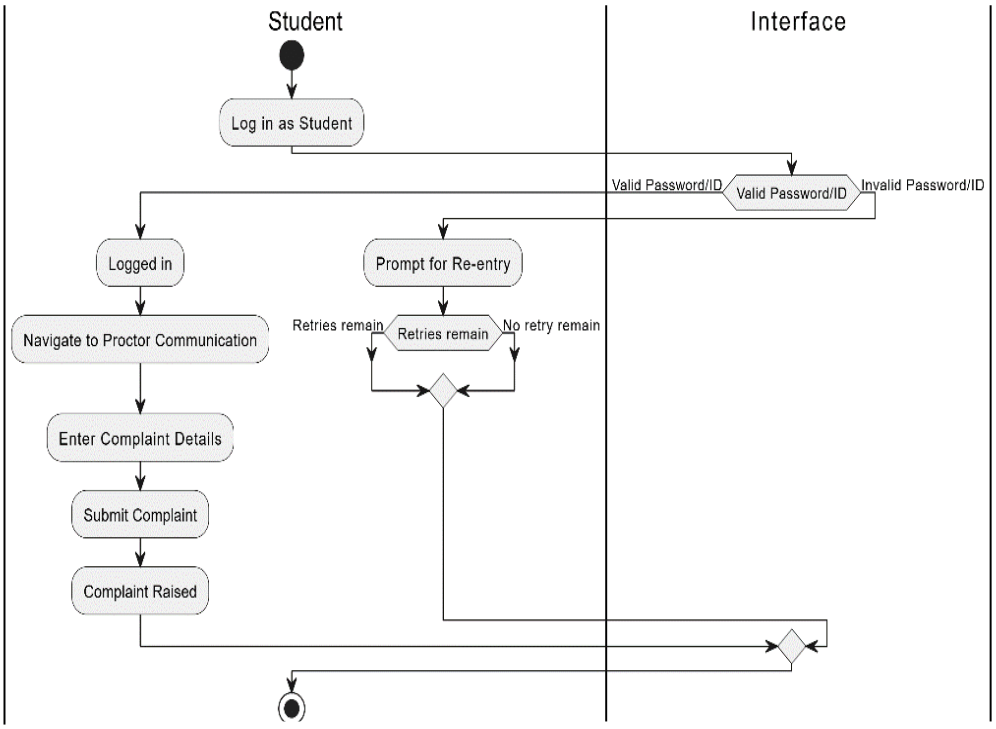


Fig: Swimlane Diagram(Raise Complaint / Issue)

Use case 12: View & Respond to Student Issues

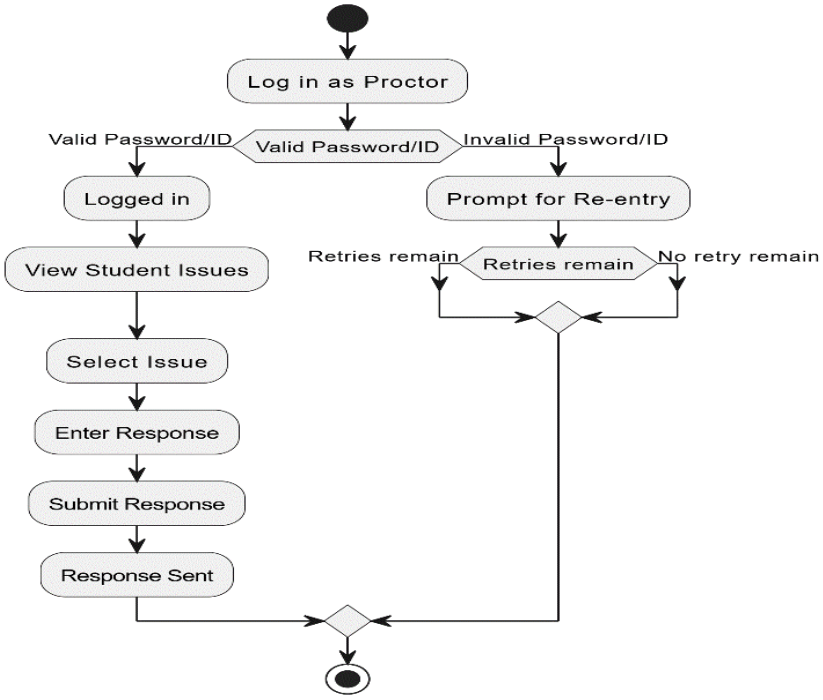


Fig: Activity Diagram(View & Respond to Student Issues)

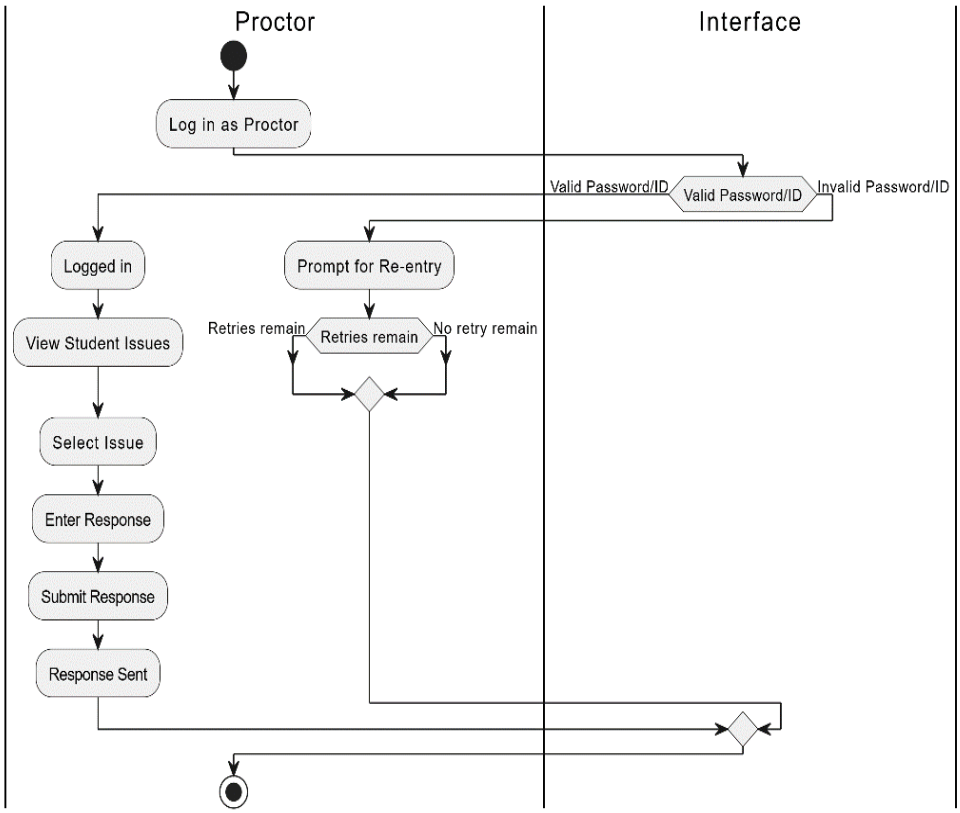


Fig: Swimlane Diagram(View & Respond to Student Issues)

Use case 13: Mark Issue as Resolved

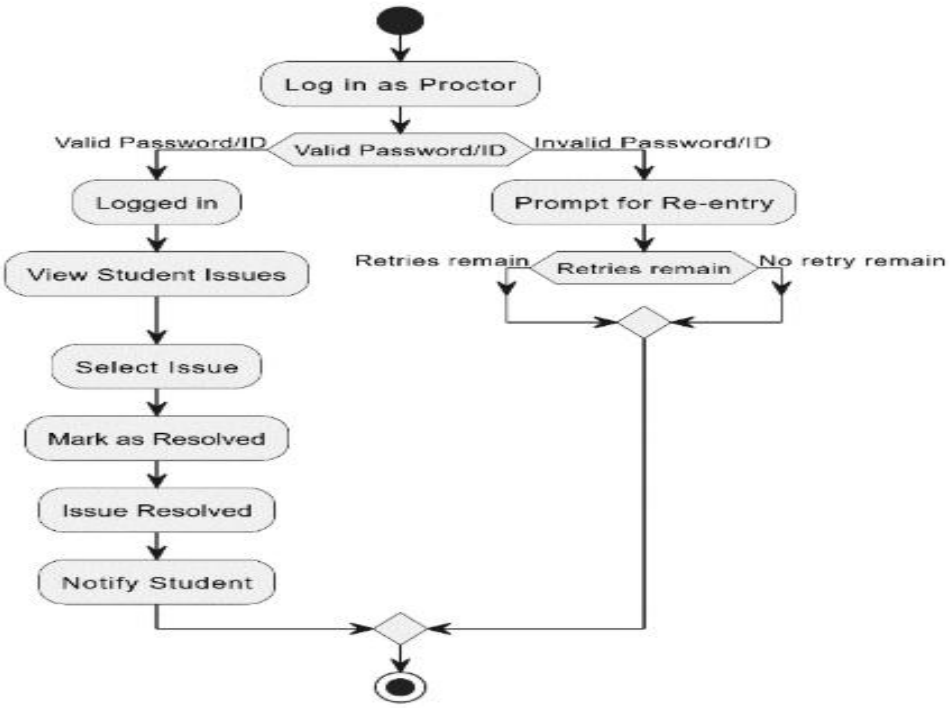


Fig: Activity Diagram(Mark Issue as Resolved)

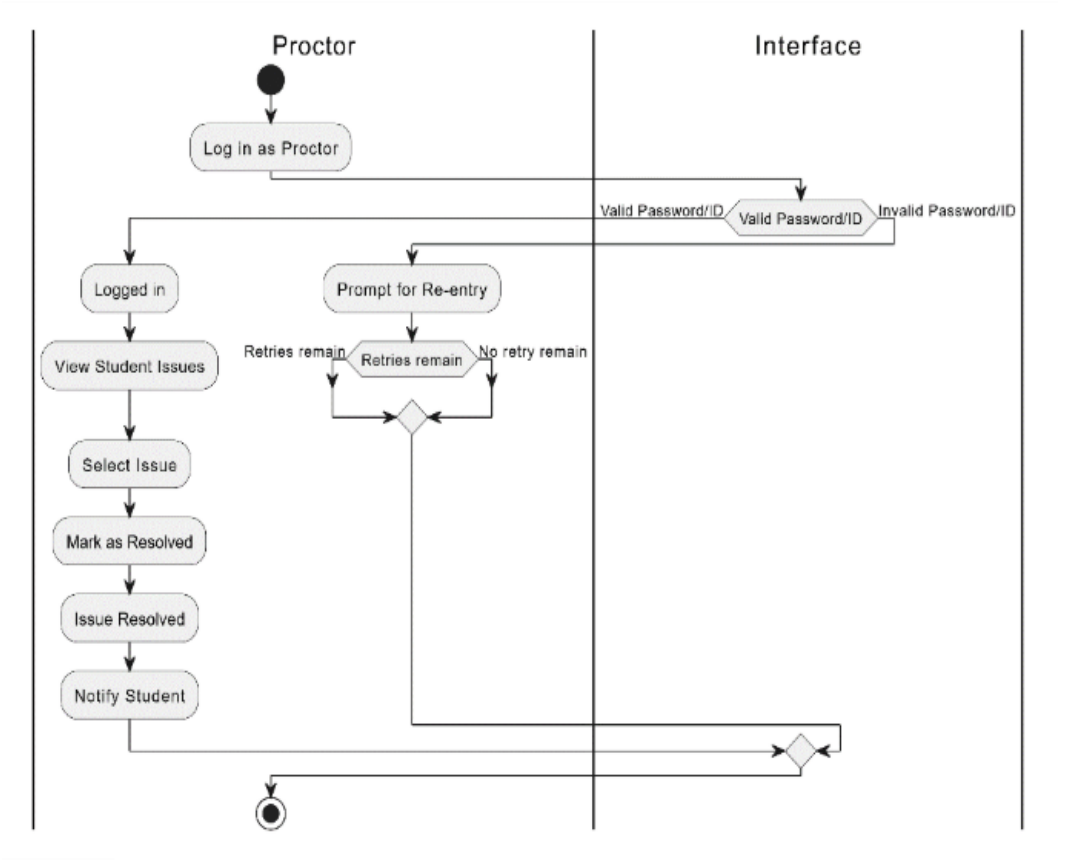


Fig: Swimlane Diagram(Mark Issue as Resolved)

Use case 14: View Room Availability Calendar

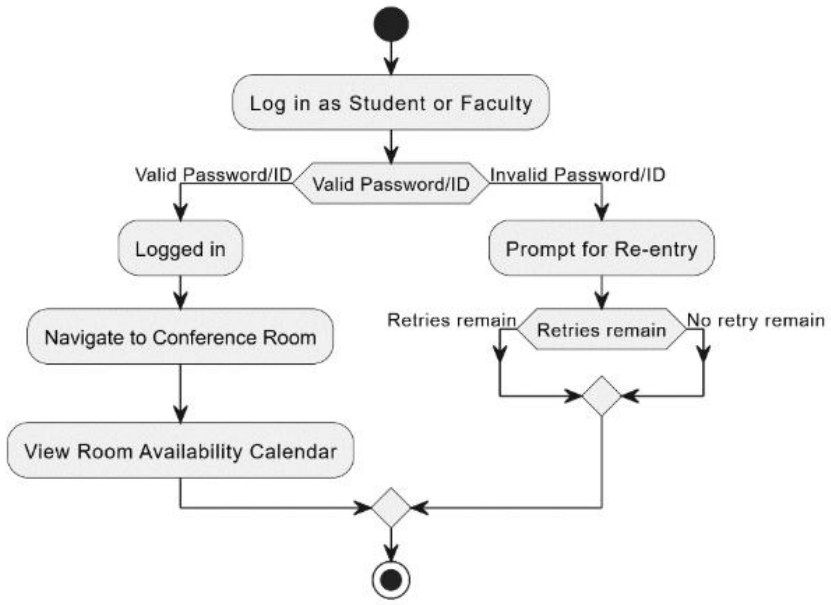


Fig: Activity Diagram(View Room Availability Calendar)

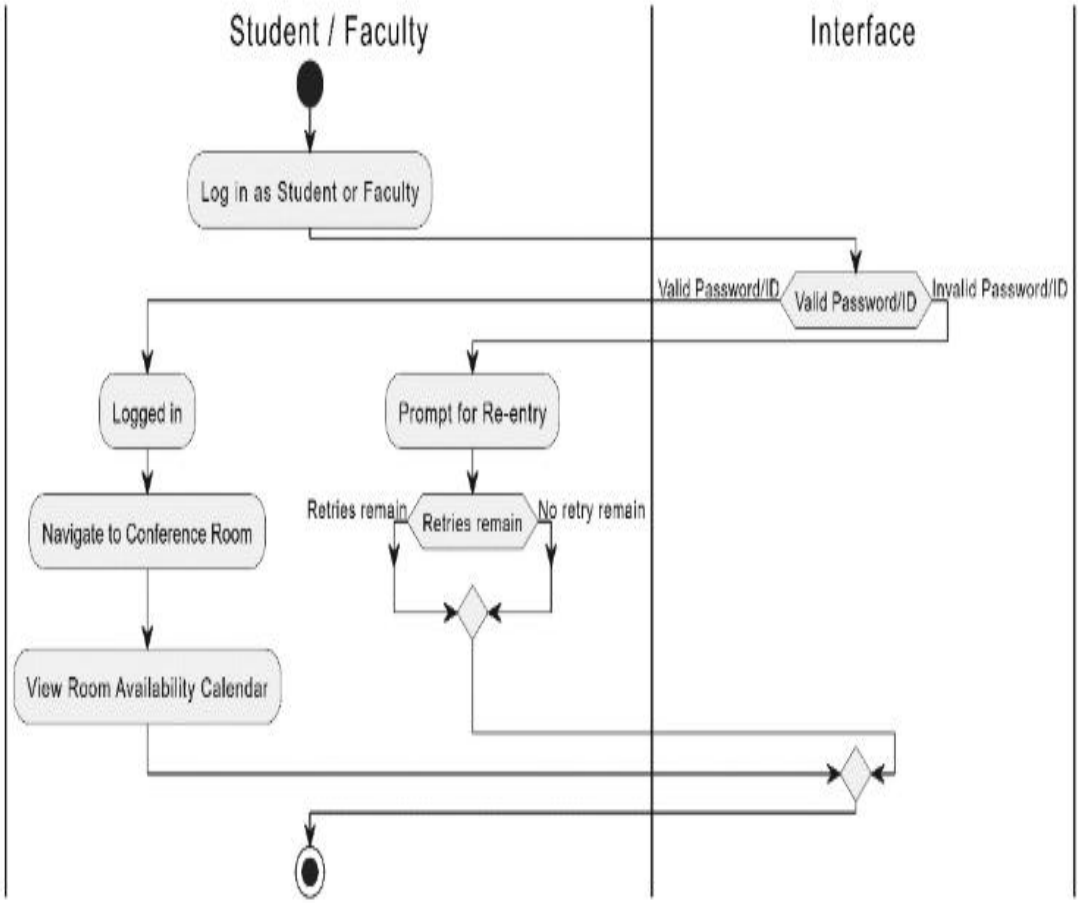


Fig: Swimlane Diagram(View Room Availability Calendar)

Use case 15: Submit or Cancel Request (Conference Room)

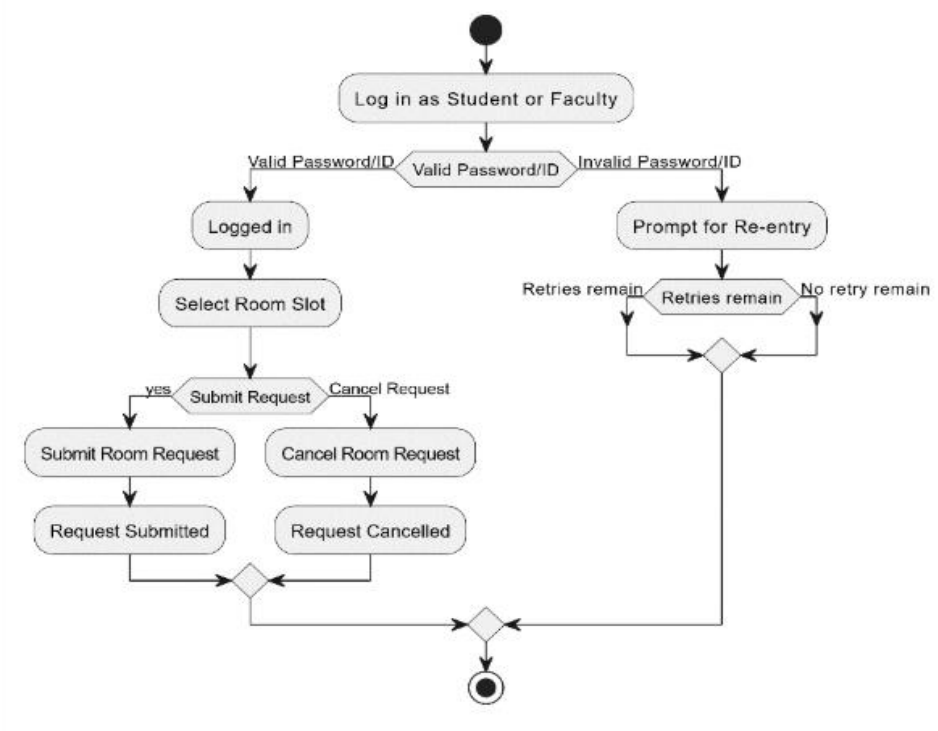


Fig: Activity Diagram(Submit or Cancel Request (Conference Room)

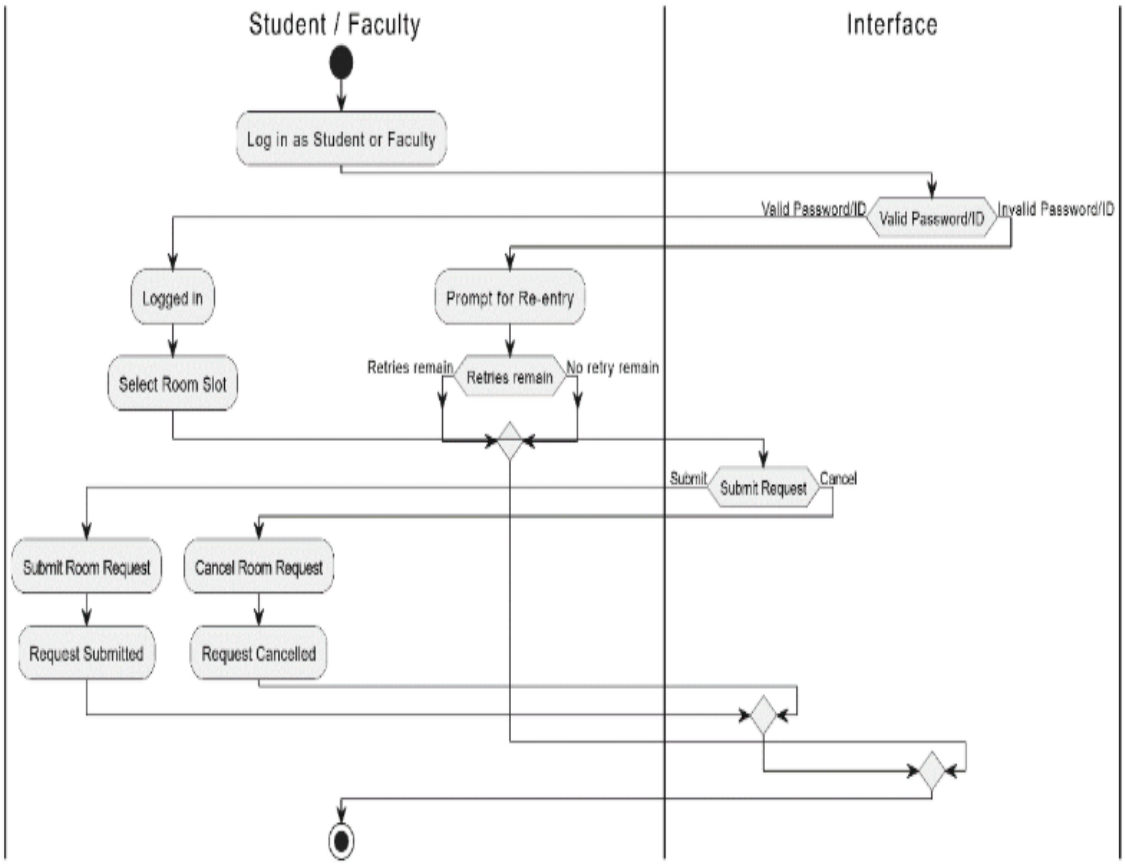


Fig: Swimlane Diagram(Submit or Cancel Request (Conference Room)

6.1 Entity Relation (ER) Diagram

Step 1: Identified Entities

The following entities are identified from the system model:

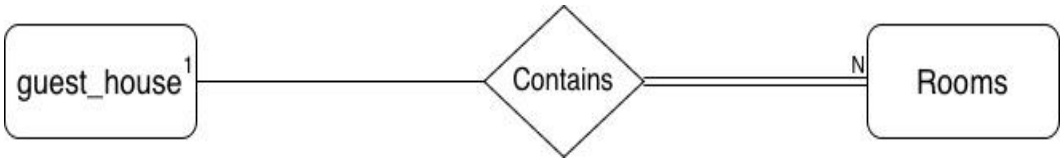
1. User
2. Rooms
3. Beds
4. Guest_house
5. Bookings
6. Conference_rooms
7. Conference_bookings
8. Sports_fields
9. Sports_bookings

Step 2: Attributes and Primary keys

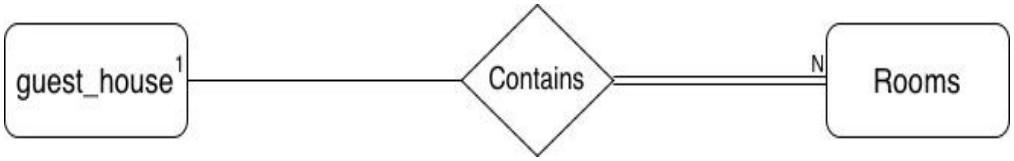
1. User: { id, name, role, email, password, designation, department, gender }
2. Rooms: { id, guest_house_id (FK), room_number, type, status }
3. Beds: { id, room_id (FK), bed_number, is_booked }
4. Guest_house: { id, name, district, description, total_rooms, abailable_rooms }
5. Bookings: { id, user_id(FK), room_id(FK), status, check_in, check_out }
6. Conference_rooms: { id, name, capacity, location }
7. Conference_bookings: { id, user_id (FK), con_id (FK), status, start_time, end_time }
8. Sports_fields: {id, name, location }
9. Sports_booking: { id, user_id (FK), field_id (FK), booking_date, slot, purpose, team_name, participants, notes, status, rejection_reason }

Step 3: Cardinality and Relationship

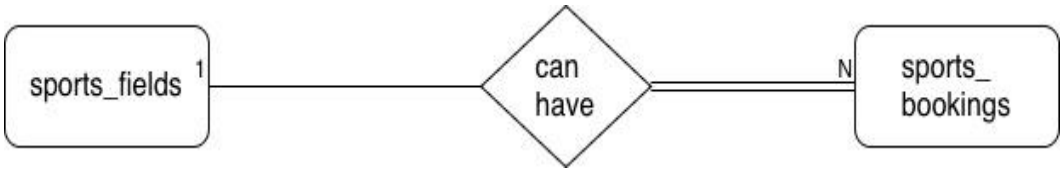
1. **Guest_house --- Rooms**
   Cardinality: one-to-many (1:M)



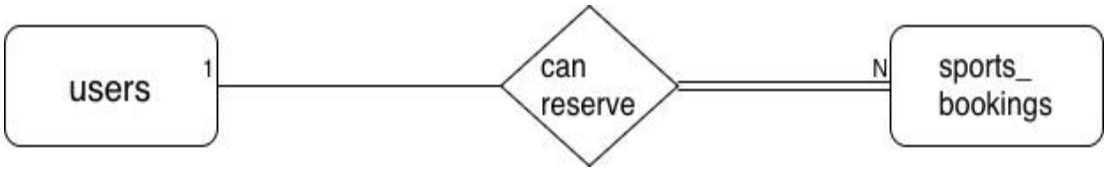
2. **Guest_house --- rooms**
   Cardinality: one-to-many (1:M)



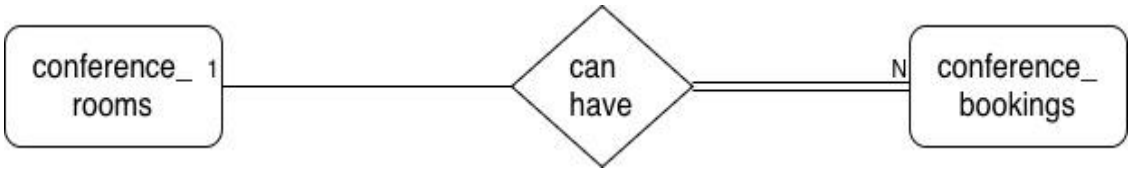
3. **Sports_fields --- Sports_bookings**
   Cardinality: one-to-many (1:M)



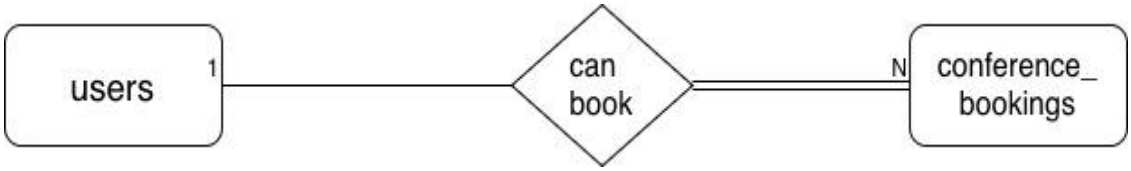
4. **Users --- Sports_bookings**
   Cardinality: one-to-many (1:M)



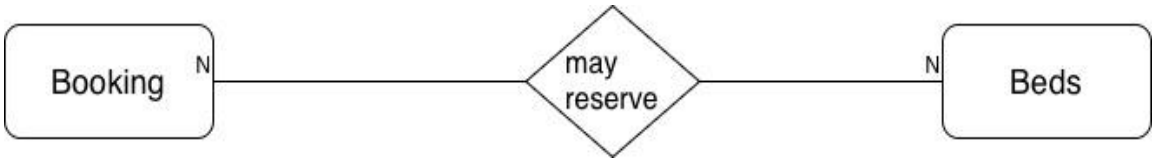
5. **Conference_room --- Conference_booking**
   Cardinality: one-to-many (1:M)



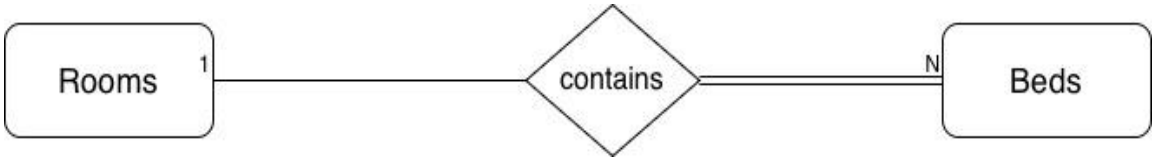
6. **User --- Conference_booking**
   Cardinality: one-to-many (1:M)



7. **Booking --- Bed**
   Cardinality: many-to-many (N:M)

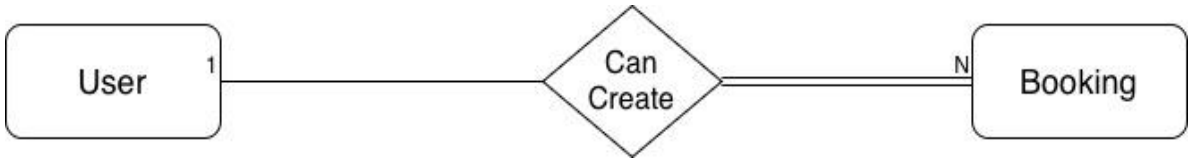


8. **Rooms --- Bed**
   C



9. **User --- Booking**

   Cardinality: one-to-many (1:M)



Step 4: ER Diagram

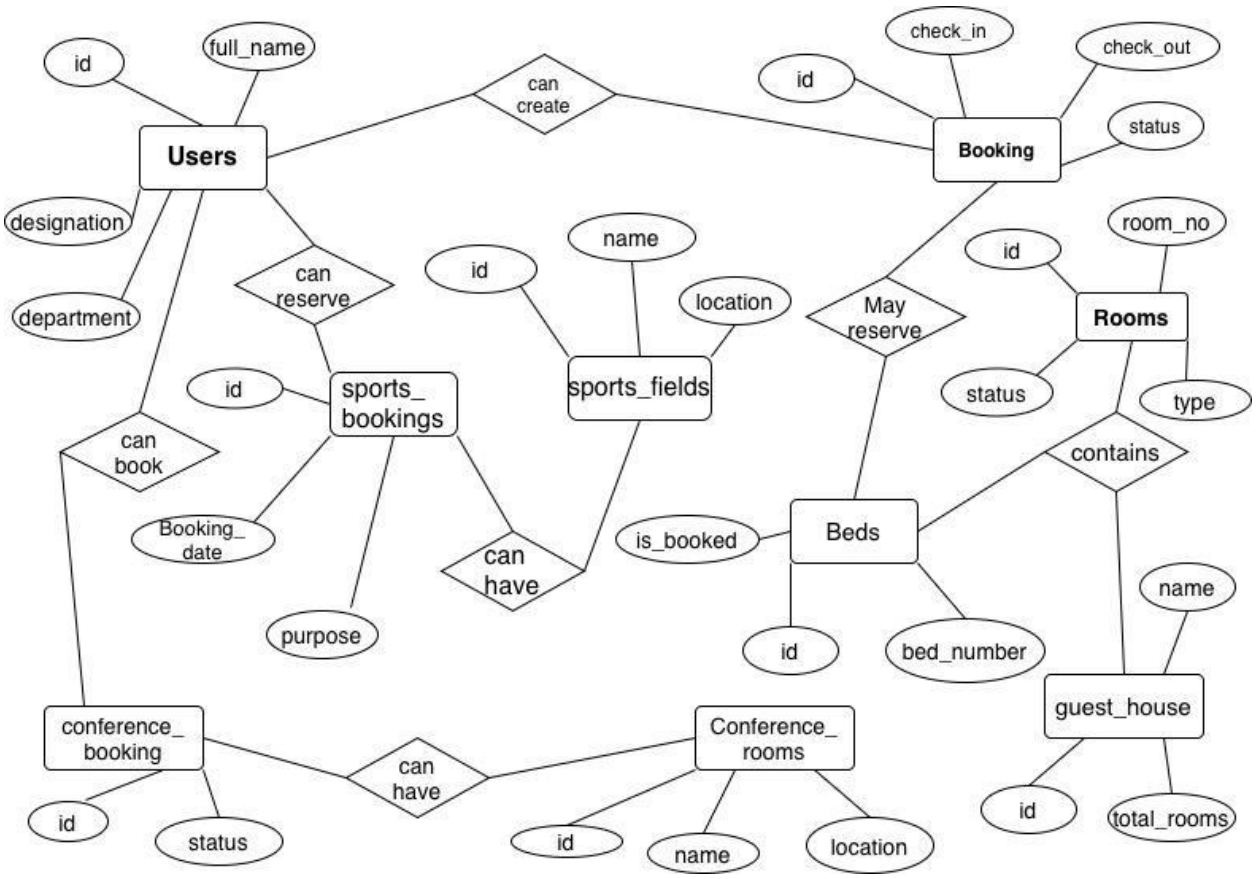


Fig: ER-Diagram

Step 5: Schema Diagram

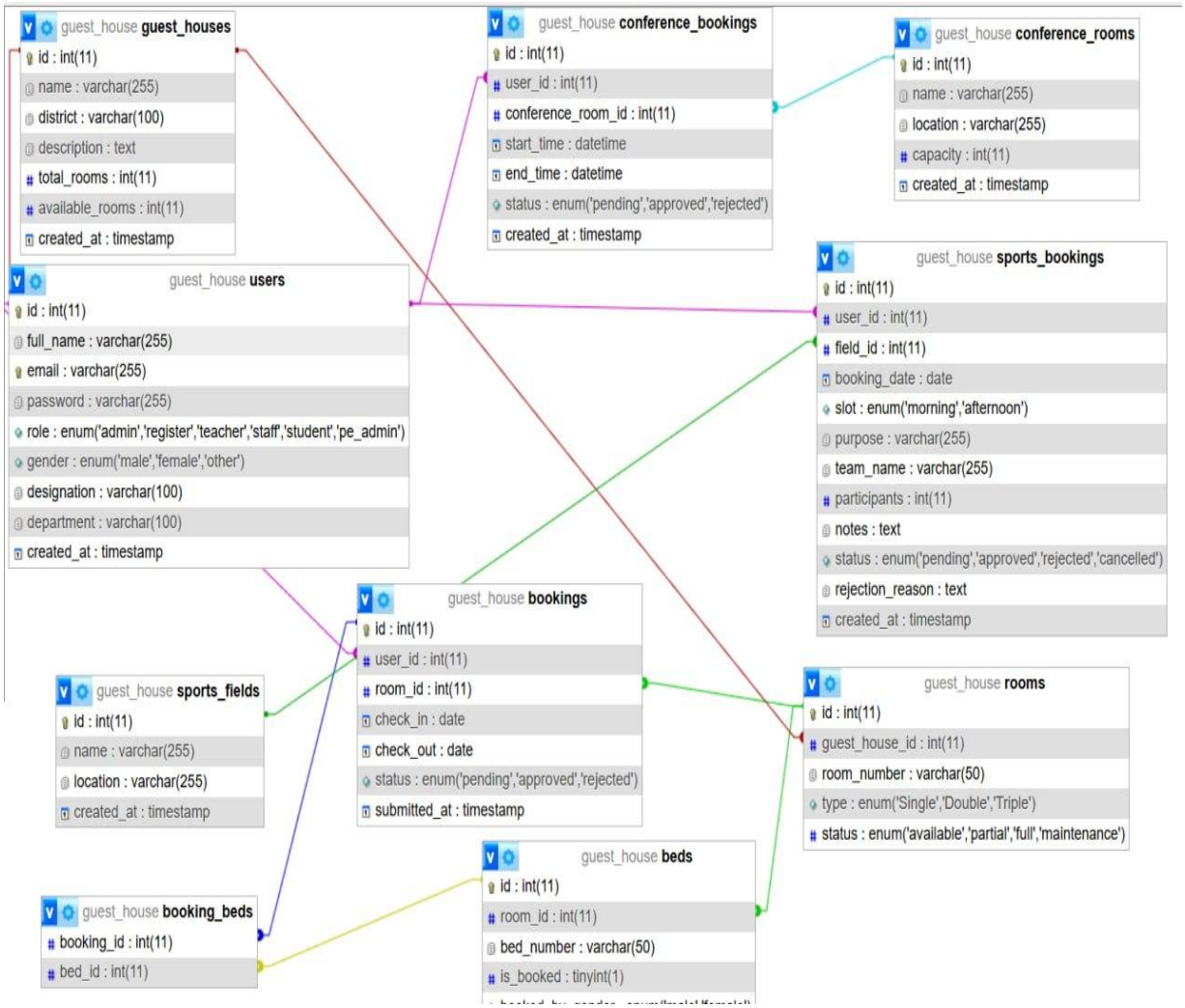


Fig: Schema Diagram

7.Class-Based Modeling

7.1.Identify Class and Attributes:

Step-1: Identifying and categorize all nouns

External Entities	Student, Faculty, Teacher, Authority, Admin Staff, Users
Things	Booking, Request, Appointment, Application, Record, Status, Notification
Event	Submit, Reserve, Review, Reject, Approve, Notify
Roles	Admin, Proctor, Students Affairs Office
Place	University, Conference Room, Central Field, Guest House
Structure	Database, System, Web Platform

Step-2: Selection of potential class

Selection characteristics:

- Retained information
- Needed services
- Multiple attributes
- Common attributes
- Common operations
- Essential requirements

Potential Class	Characteristic Number That Applies
Users	Accepted
Faculty	Rejected: 3 fails
Teacher	Rejected:1, 3 fails
Authority	Rejected: 3,5 fails
Admin Staff	Rejected: 3,5 fails
Student	Rejected: 3,4,5 fails
Booking	Rejected: 3 fails

Potential Class	Characteristic Number That Applies
Appointment	Rejected: 3 fails
Application	Rejected: 1,3,5 fails
Record	Rejected: 3 fails
Status	Rejected: 3 fails
Notification	Rejected: 1,3,5 fails
Submit	Rejected: 1,3,5 fails
Reserve	Rejected: 3 fails
Review	Rejected: 1,3,5 fails
Reject	Rejected: 3 fails
Approve	Rejected: 3 fails
Notify	Rejected: 3 fails
Admin	Accepted
Proctor	Rejected: 3,5 fails
Students Affairs Office	Rejected: 3,5 fails
University	Rejected: 1,3,5 fails
Conference Room	Accepted
Central Field	Accepted
Guest House	Accepted
Database	Accepted
System	Accepted

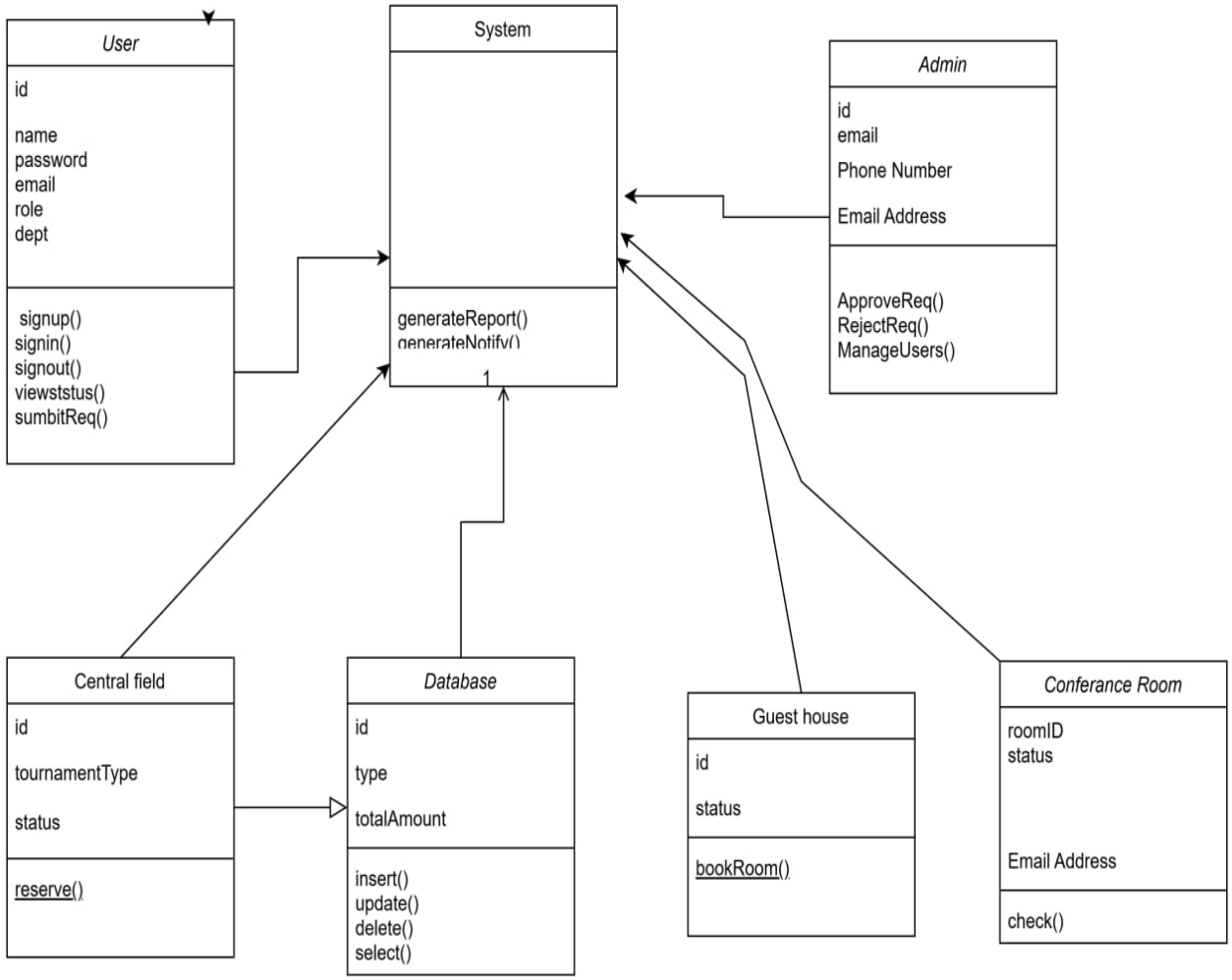


Fig:Class Based Model

8.Flow Oriented Modeling

8.1.Data Flow Diagram:

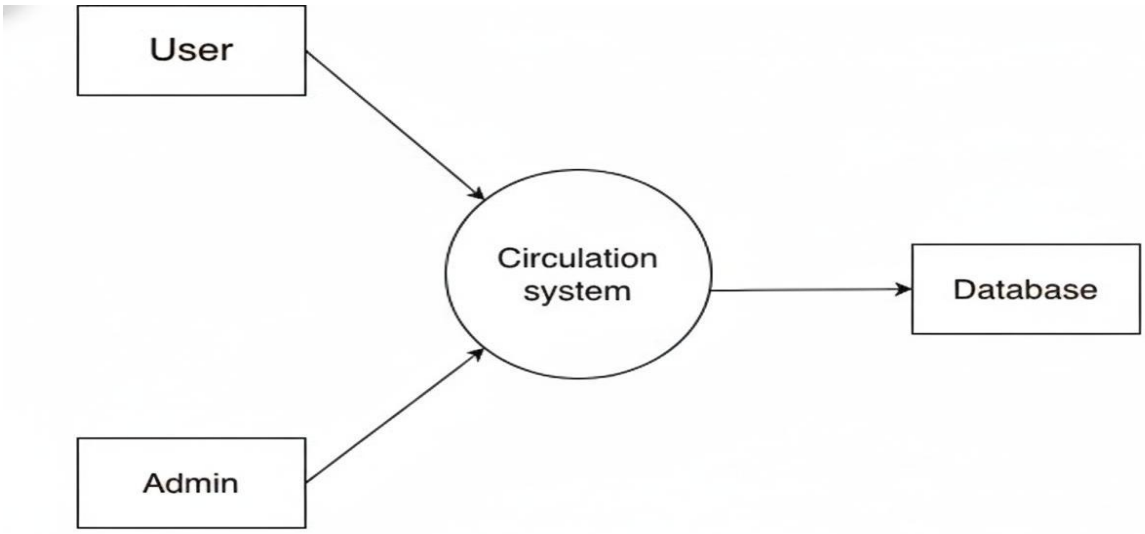


Fig: Level-0 For CirculationSystem

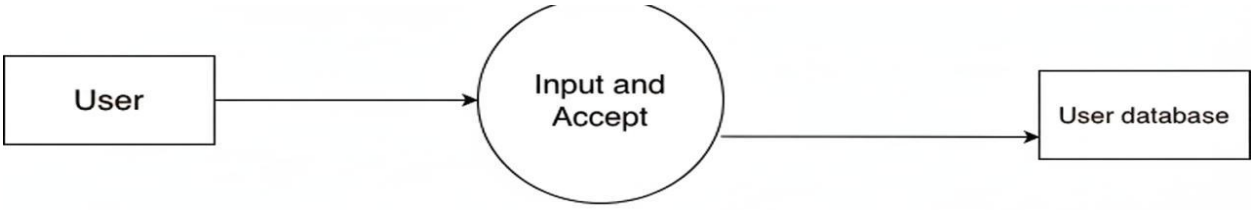


Fig: Level-1.1 For CirculationSystem(User)

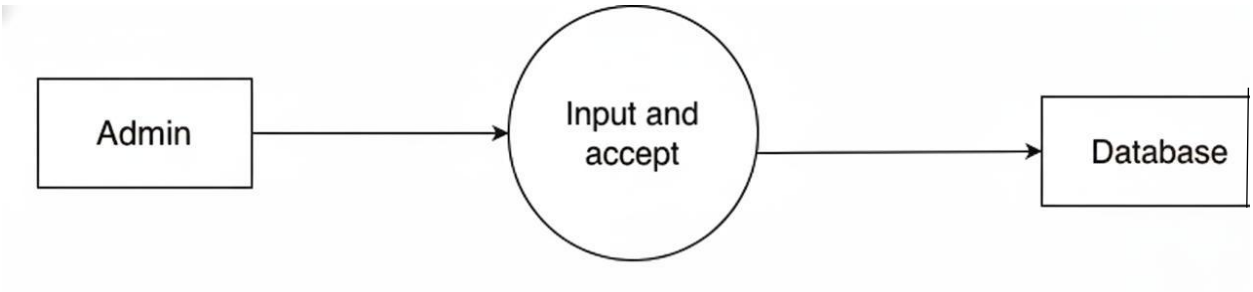


Fig: Level-1.2 For CirculationSystem(Admin)

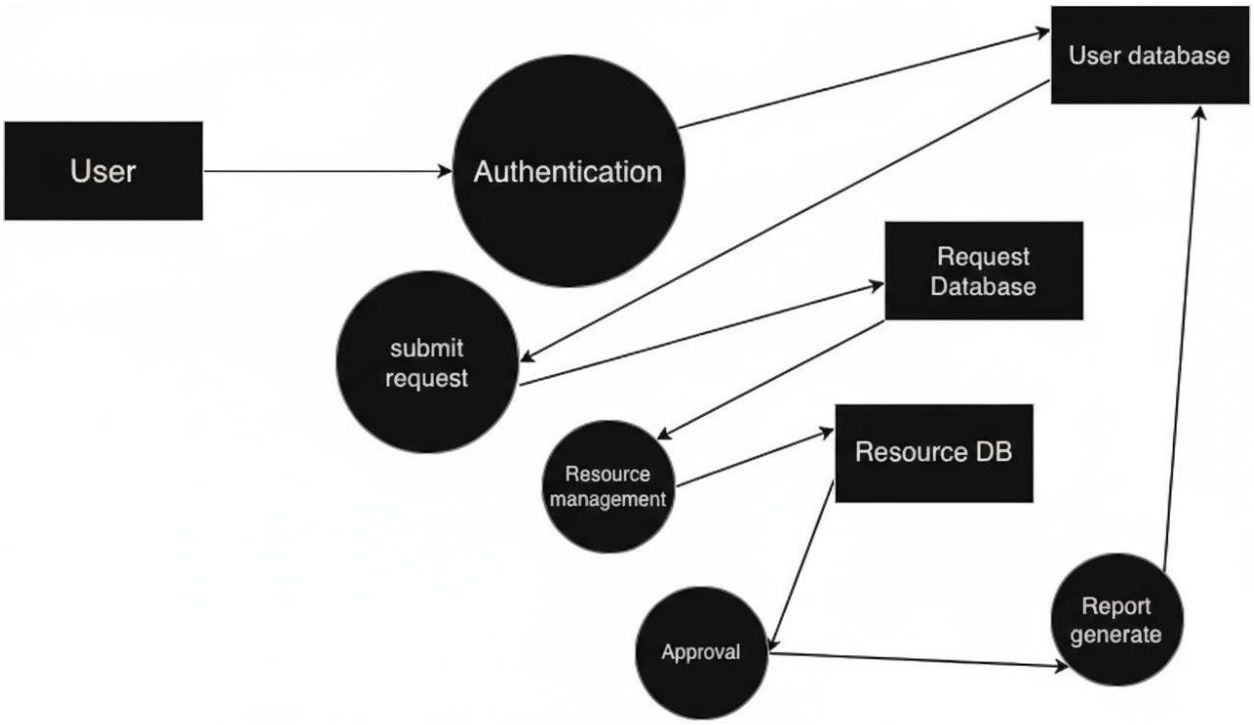


Fig: Level-2.1 For CirculationSystem(User)

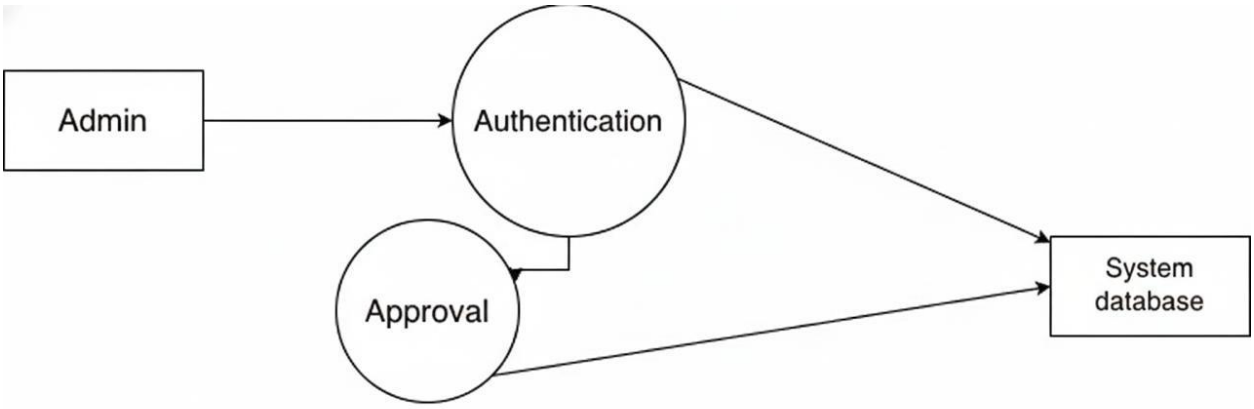


Fig: Level-2.2 For CirculationSystem(Admin)

9.Behavioral Model

9.1.Sequence Diagram:

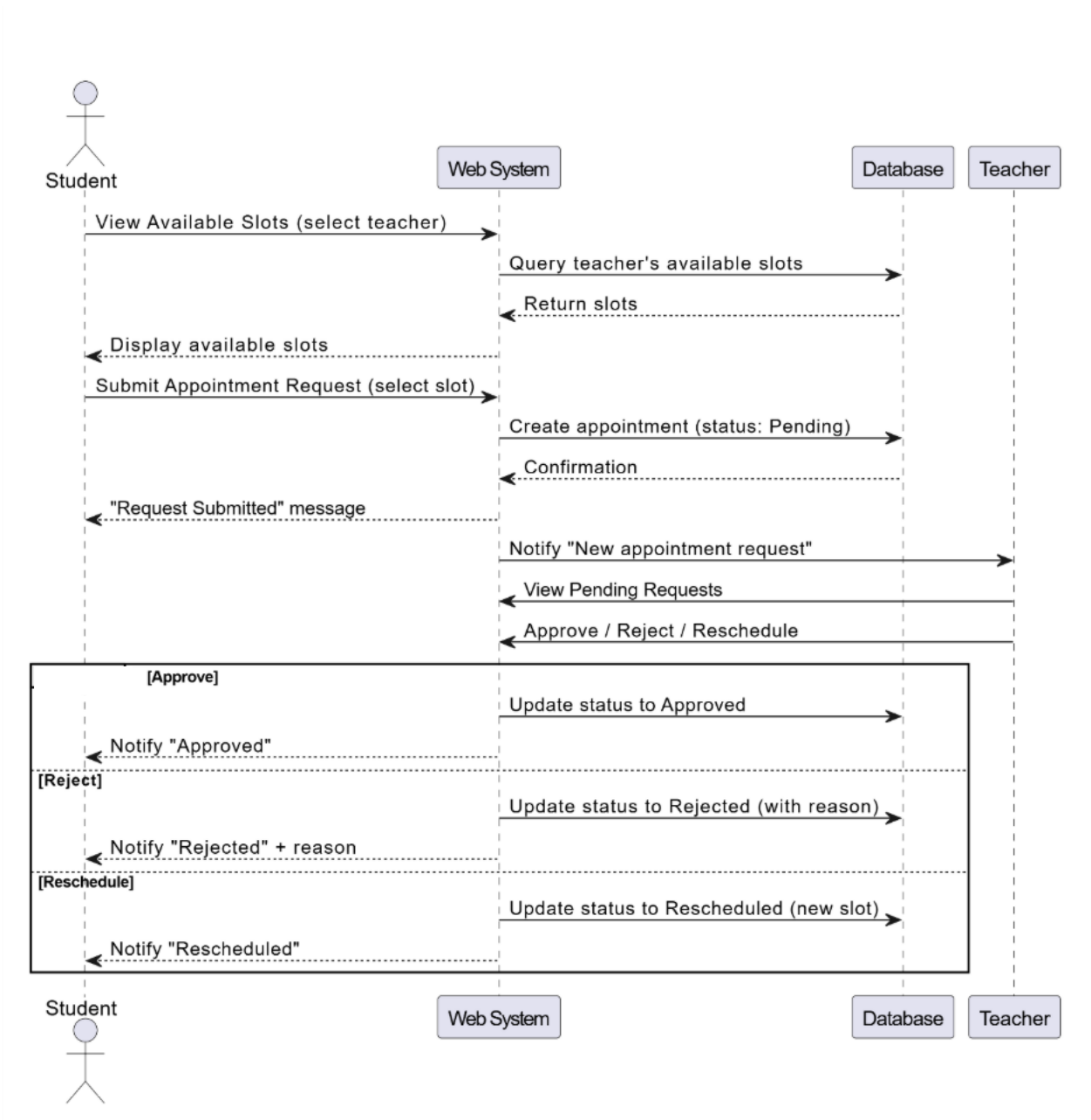


Fig: Teacher Appointment Process

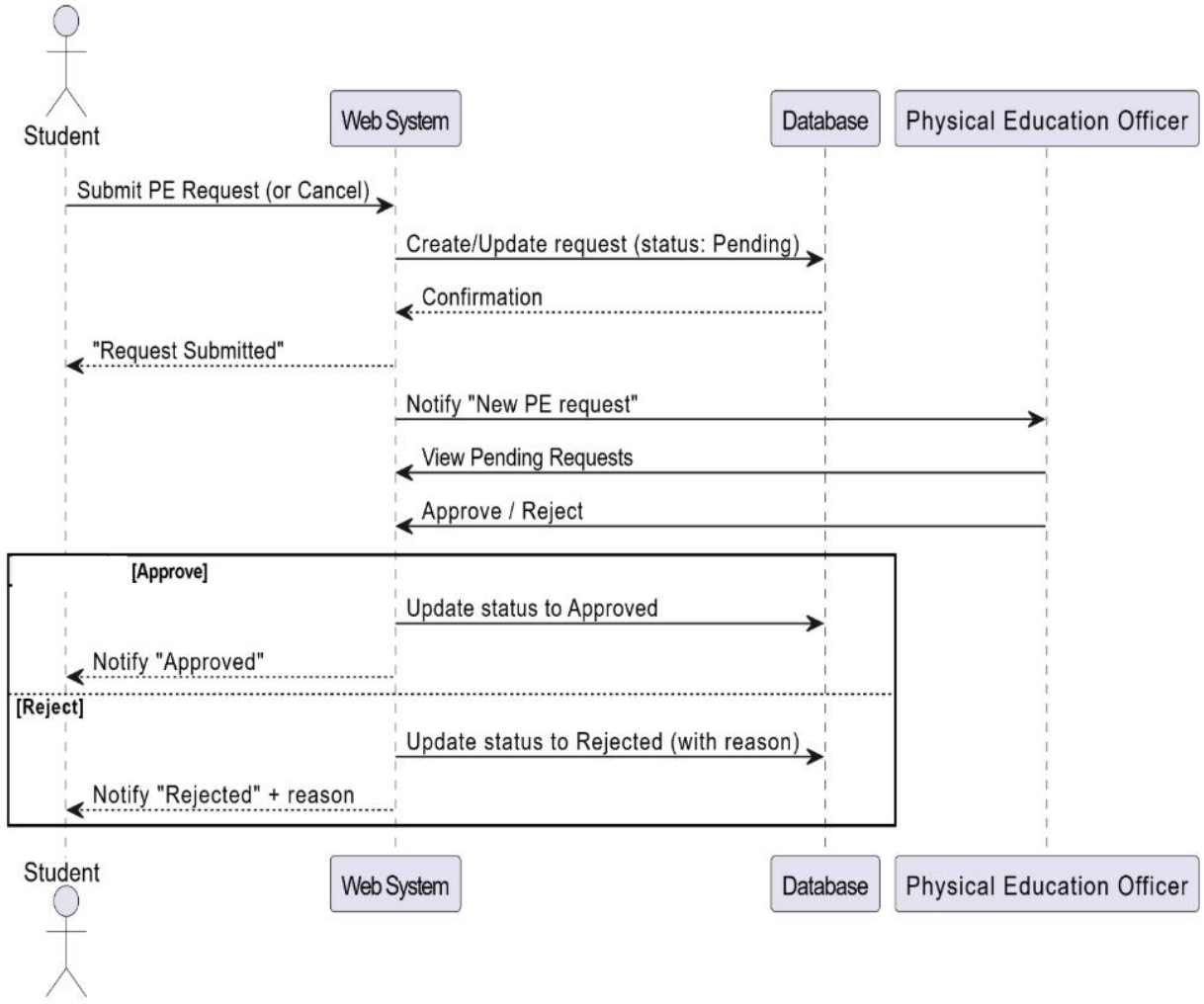


Fig: Physical Education Request

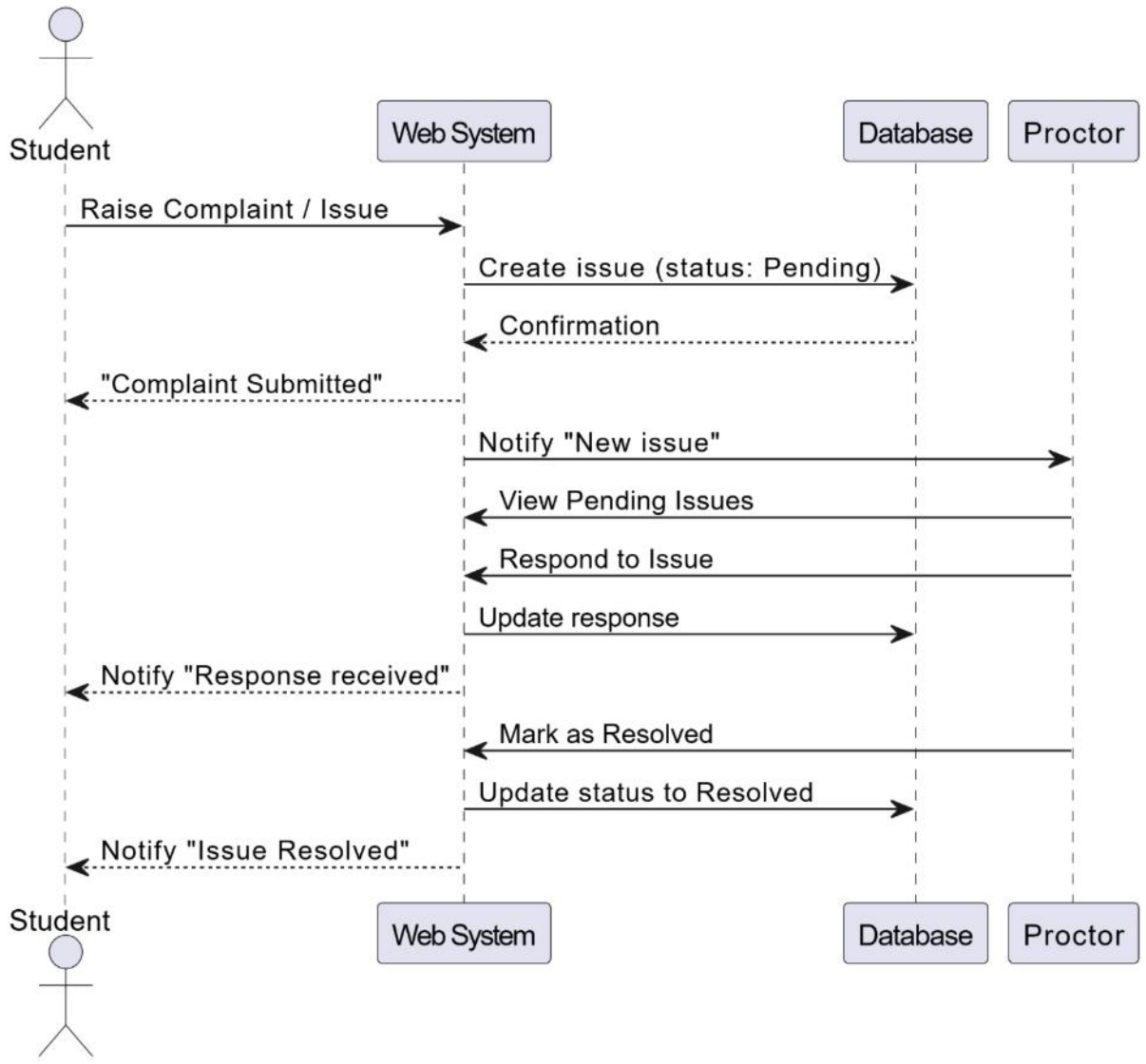


Fig: Proctor Complaint Process

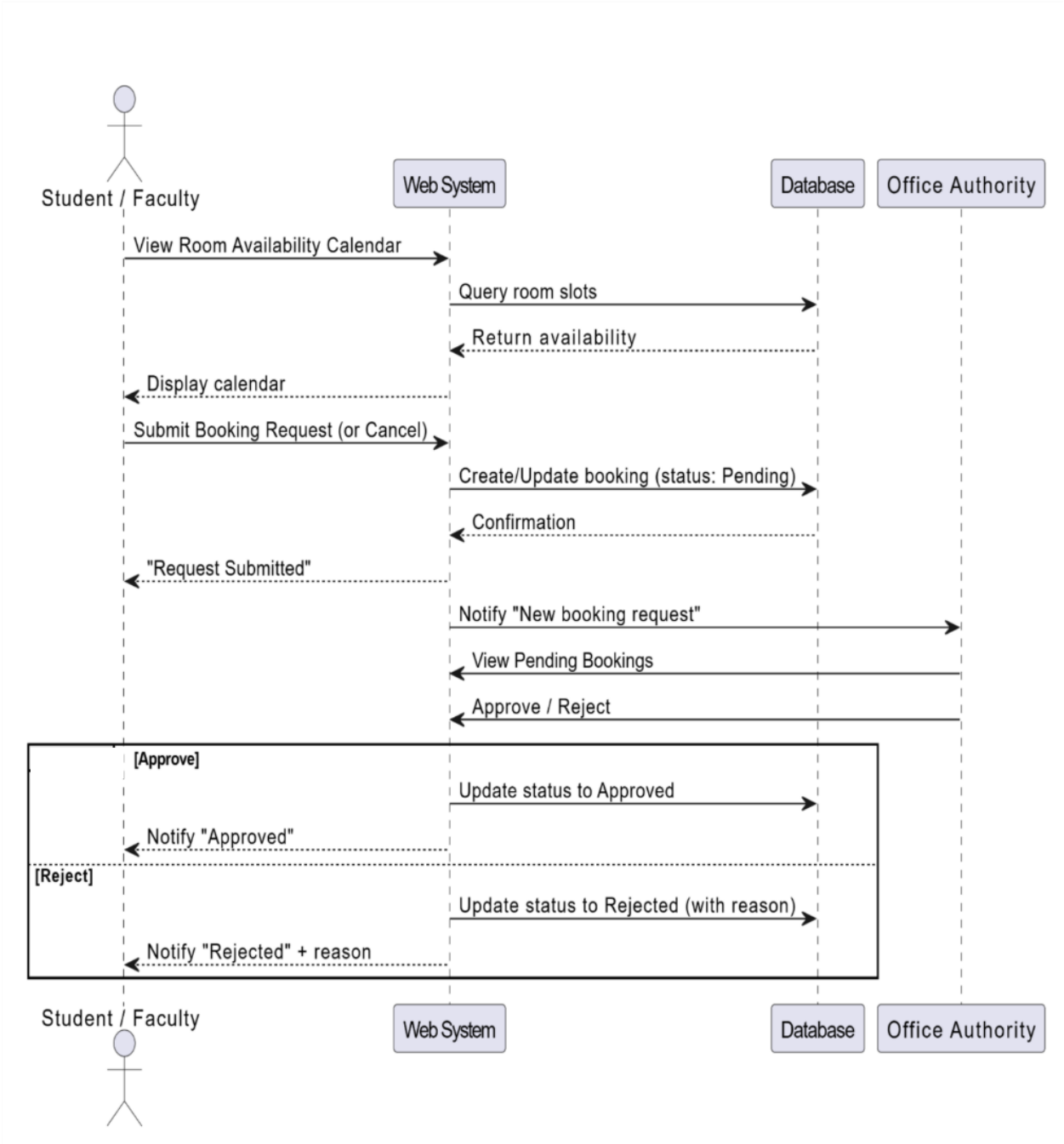


Fig: Conference Room Booking

9.2.State Diagram:

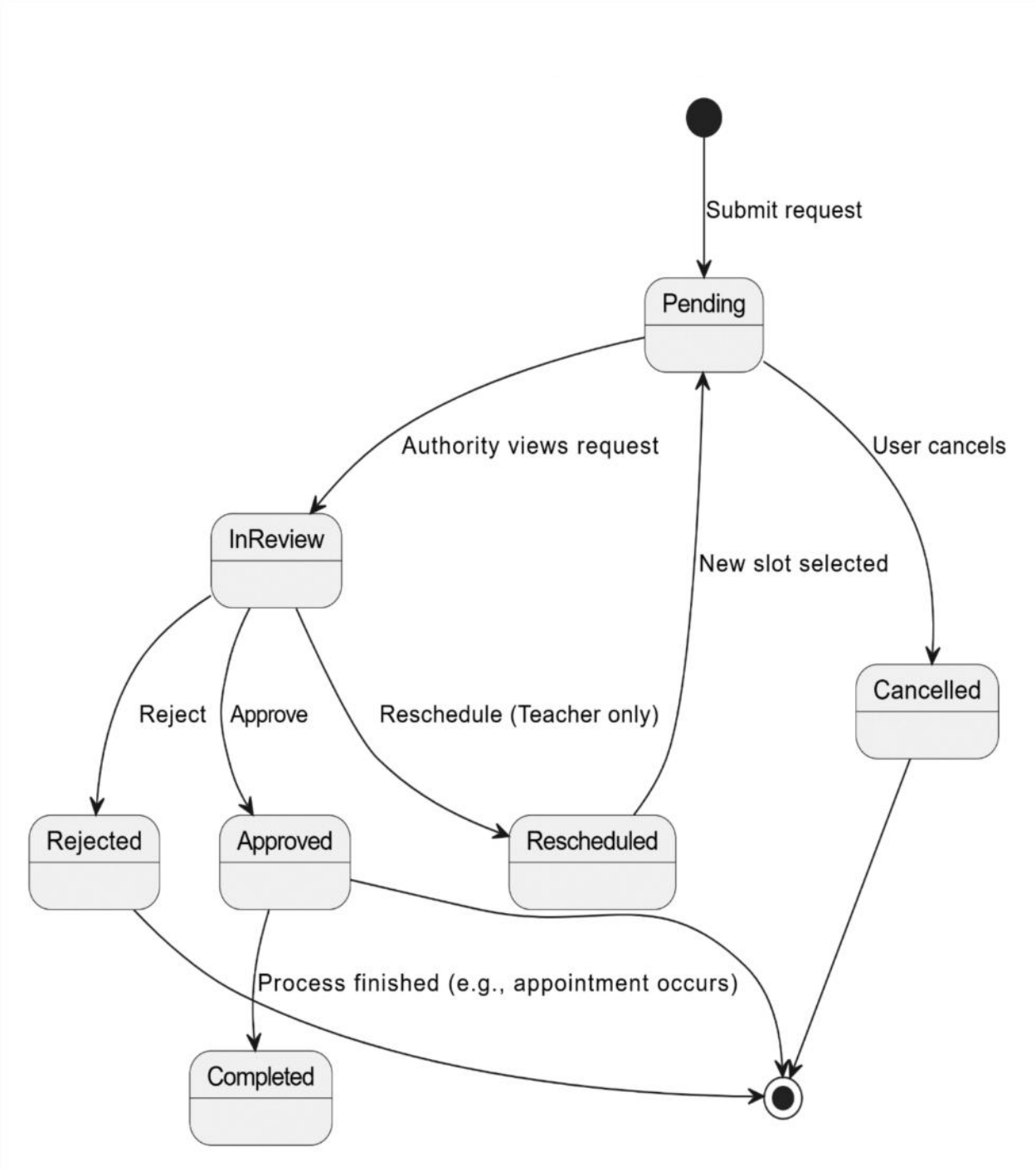


Fig: Appointment/Request/Booking Life Cycle

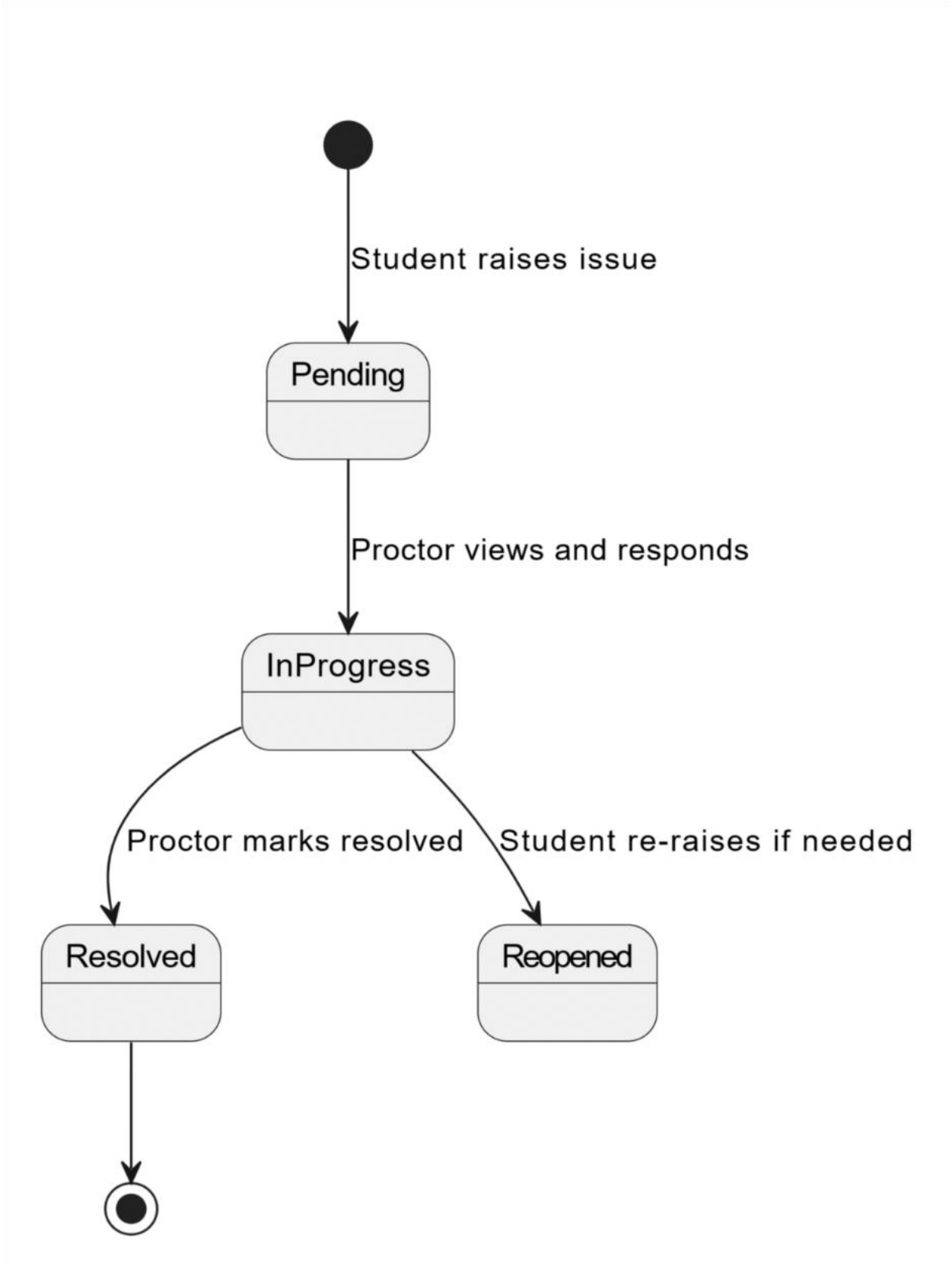


Fig: Complaint and Issue LifeCycle

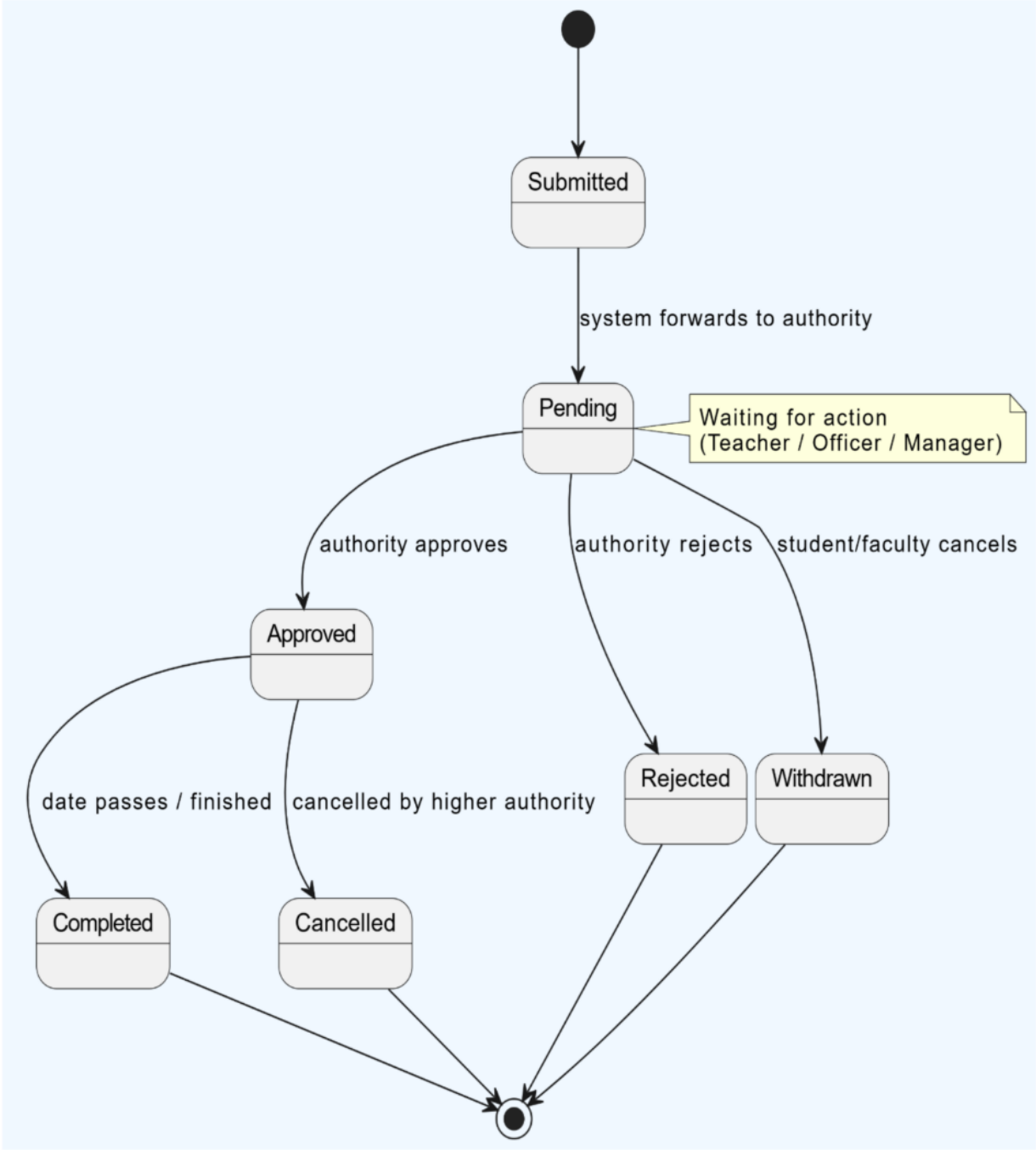


Fig: Accept & Reject Life Cylce

THE END